Positive Certificates and Primitive Arithmetic for Euler's Constant

Target realization, exact optimization, and arithmetic obstructions

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A consolidated research manuscript with complete proofs

Abstract

We develop a unified framework for positive, logarithm-free approximation to Euler's constant in which every estimate is compared with the actual reduced denominator. Canonical Binet kernels yield rational brackets and a least-denominator selection with an exact rationality dichotomy. Farey geometry reduces the primitive-error minimum on a rational interval to logarithmically many candidates. Global positive target realization, effective polynomial-square representations, optimal Laguerre integerization, prime-residue tests, and support compression provide complementary analytic and arithmetic tools. We prove divergence for entire canonical, fixed-polynomial, and dyadic interpolation classes. A localized positive family has the exact arithmetic threshold $q_M = o(M)$, and a one-step cutoff change removes a specified adverse prime factor from its reduced denominator. Signed cancellation gives an exponentially small integerized upper bound but does not guarantee nonvanishing. Additional synthesis results determine the sharp asymptotic degree of positive dyadic certificates, extend the selection dichotomy to general shrinking rational intervals, exhibit an unbounded gap between least-denominator and primitive-optimal selection, and prove an exponential height barrier for independent signed outputs. All statements retain their explicit hypotheses. The resulting tools do not establish the irrationality of Euler's constant.

Keywords. Euler's constant; rational approximation; positive kernels; Farey fractions; Laguerre polynomials; integer linear forms.

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1 The objective and the ten result groups

Throughout,

$$\gamma = \lim_{n\to\infty}(H_n - \log n), \qquad H_n = \sum_{j=1}^{n} \frac{1}{j}, \qquad H_0 = 0.$$

For a reduced rational p/q, $q > 0$, its *primitive error* is

$$\mathcal{E}(p/q) = q\left|\gamma - \frac{p}{q}\right| = |q\gamma - p|.$$

The coefficients are integers; the value itself is not asserted to be an integer. If $\gamma = a/b$ in lowest terms, then $b(q\gamma - p) = aq - bp \in \mathbb{Z}$. Consequently a proved infinite sequence satisfying

$$0 < |q_j\gamma - p_j| \longrightarrow 0, \qquad p_j, q_j \in \mathbb{Z},$$

would establish irrationality. The five requirements are exact logarithm cancellation and integer coefficients, nonvanishing, decay after actual denominator reduction, an infinite family, and unconditional proofs of these assertions for the same family.

This manuscript consolidates the ten result groups selected from three research notes [1, 2, 3]. Their constructions and proofs are reproduced and connected here, so no separate manuscript or supplement is needed to follow the argument. Standard external theorems are stated when used and attributed in the bibliography. The order below is a dependency order; the first column records the earlier research priority ranking, not a claim of historical priority.

Rank	Result group	Principal statements
1	Explicit least-denominator selection and its rationality dichotomy	Theorems 5.3, 5.2.
2	Exact finite primitive-error optimization and certificates	Theorems 6.1, 6.2, 6.3.
3	Constructive positive target realization and complete reduction	Theorems 2.3, 4.3.
4	Complete polynomial-square target image and effective degree	Theorems 9.1, 9.2.
5	Optimal Laguerre integerization and surviving-prime tests	Theorems 10.1, 10.2.
6	Primitive-cost obstructions for entire coefficient classes	Theorems 8.1, 10.3, 8.4.
7	Support compression with exact denominator and error transfer	Theorem 11.1.
8	Positive rational localization and the exact threshold	Theorems 12.1, 12.2.
9	Removal of an adverse prime factor by changing the cutoff	Theorem 12.3.
10	Integerized signed cancellation and its nonvanishing obstruction	Theorem 13.2.

What is additionally proved in this consolidation. The synthesis sections prove a sharp degree law for globally nonnegative dyadic kernels, a general interval-selection dichotomy with a Fibonacci counterexample to asymptotic optimality of least-denominator selection, and a lower bound for the height of two independent outputs in the actual signed-cancellation lattice. These conclusions are proved here without assuming the irrationality of γ. Their relation to classical polynomial and Diophantine tools is made explicit; no literature priority or external peer-review status is asserted.

Conventions. All logarithms are natural unless a base is shown. The notation v_p is the usual rational p-adic valuation, and $\mathrm{odd}(n)$ is the largest odd divisor of a nonzero integer n, with its sign discarded. Empty products are one. A polynomial's coefficient height always refers to the explicitly stated basis. Different uses of A, B and of q are locally specified: the polynomial-square forms $A(f), B(f)$ are distinct from canonical coefficient normalizers, and a polynomial variable is distinguished from a reduced denominator by its argument.

2 Canonical compressed families with exact positive kernels

This section analyzes a complete low-degree family inside the shared-cutoff construction. No irrationality claim is made. The new construction gives positive, strictly monotone rational approximants in two parameters; its primitive cost can also be settled uniformly.

Write

$$d\mu(u) = \frac{du}{e^{2\pi\sqrt{u}} - 1}, \qquad \mu_j = \int_0^\infty u^j \, d\mu(u) = \frac{|B_{2j+2}|}{2j+2}.$$

For an integer $n \geq 1$, Binet's identity [4] is

$$H_n - \frac{1}{2n} = \gamma + \log n - \Phi(n^2), \qquad \Phi(z) = \int_0^\infty \frac{d\mu(u)}{u+z}.$$

An elementary moment estimate used below is

$$\frac{\mu_{j+1}}{\mu_j} = \frac{(2j+3)(2j+2)}{4\pi^2} \frac{\zeta(2j+4)}{\zeta(2j+2)} < \frac{(j+1)^2}{4}. \tag{1}$$

Indeed the zeta ratio is less than 1, $\pi > 3$, and $(2j+3)(2j+2) < 9(j+1)^2$.

Theorem 2.1 (An exact positive and doubly monotone canonical family)**.** *For integers $m, t \geq 0$, put*

$$P_m(x) = \prod_{j=1}^{m} (4^j x - 1), \quad A_m = P_m(1), \quad s_m = \frac{P_m'(1)}{P_m(1)}, \quad a = 2^t, \quad b = t + s_m, \quad R = 4^{m+1}.$$

Empty products have value 1, so $s_0 = 0$. Define

$$C_{m,t}(x) = \frac{P_m(x)}{A_m}[1 - b(x-1)] = \sum_i c_i x^i, \qquad r_{m,t} = \sum_i c_i \left(H_{a2^i} - \frac{1}{2a2^i} \right).$$

Then $r_{m,t}$ is rational and

$$E_{m,t} := \gamma - r_{m,t} = \int_0^\infty \frac{u^m \{u + a^2[R + b(R-1)]\}}{\prod_{i=0}^{m+1}(u + a^2 4^i)} \, d\mu(u) > 0. \tag{2}$$

Both coordinate increases are strict:

$$r_{m+1,t} > r_{m,t}, \qquad r_{m,t+1} > r_{m,t}.$$

Moreover, uniformly in both parameters,

$$\frac{a^{-2m-2} 4^{-m(m+1)/2}}{3^{15}} \leq E_{m,t} \leq \mu_m (t+m+3) a^{-2m-2} 4^{-m(m+1)/2}. \tag{3}$$

In particular $r_{m,t} \to \gamma$ along every sequence $m + t \to \infty$.

Proof. The defining polynomial satisfies

$$C_{m,t}(1) = 1, \quad C'_{m,t}(1) = -t, \quad C_{m,t}(4^{-j}) = 0 \quad (1 \le j \le m).$$

Binet's identity therefore cancels all logarithms exactly. To evaluate the remaining kernel, first observe the rational identity

$$\sum_{i=0}^{m} \frac{[x^i]P_m(x)}{u + a^2 4^i} = \frac{A_m u^m}{\prod_{i=0}^{m}(u + a^2 4^i)}. \tag{4}$$

After taking a common denominator, the numerator has degree at most m. Its first m Taylor coefficients at zero vanish because $P_m(4^{-j}) = 0$ for $1 \le j \le m$; its leading coefficient is $P_m(1) = A_m$. This proves the identity. Multiplication by x shifts every index by one. Applying this observation to $C_{m,t}$ gives (2), whose integrand is positive for $u > 0$.

For monotonicity, introduce the direction

$$J_{m,t} := \int_0^\infty \sum_i \frac{[x^i]\{P_m(x)(x-1)^2/A_m\}}{u + a^2 4^i}\, d\mu(u).$$

Two shifts of (4) give exactly

$$J_{m,t} = a^2(R-1) \int_0^\infty \frac{u^m\{4a^2(R-1) - 3u\}}{\prod_{i=0}^{m+2}(u + a^2 4^i)}\, d\mu(u). \tag{5}$$

This integral is strictly positive. Indeed, for the positive measure $u^m\, d\mu$ and the strictly decreasing positive weight $w(u) = \prod_{i=0}^{m+2}(u + a^2 4^i)^{-1}$, the weighted mean of u is at most μ_{m+1}/μ_m. This follows by integrating $(u - v)(w(u) - w(v)) \le 0$ against the product measure. By (1), that mean is less than $(m+1)^2/4$, whereas $4a^2(R-1)/3$ is larger. For instance $(m+1)^2 \le 4^m$ and $4(R-1)/3 \ge R$ suffice.

Direct polynomial subtraction now gives

$$C_{m+1,t} - C_{m,t} = -\frac{R[R + b(R-1)]}{(R-1)^2} \frac{P_m}{A_m}(x-1)^2,$$

$$xC_{m,t+1} - C_{m,t} = -(b+1)\frac{P_m}{A_m}(x-1)^2.$$

In the second identity the common external base is $a = 2^t$; the leading x accounts for the new base $2a$. Both differences of gamma errors are negative multiples of $J_{m,t} > 0$, proving strict monotonicity.

For the upper bound in (3), bound each denominator factor below by $a^2 4^i$. This gives

$$E_{m,t} \le \frac{\mu_{m+1} + a^2[R + b(R-1)]\mu_m}{a^{2m+4} 4^{(m+1)(m+2)/2}}.$$

Since $s_m = m + \sum_{j=1}^{m}(4^j - 1)^{-1} < m + 1$, and (1) implies $\mu_{m+1}/(a^2 R \mu_m) < 1$, this proves the stated upper bound.

For the lower bound, integrate only over $1 \le u \le 2$. On that interval $d\mu(u) > 3^{-12} du$, because $2\pi\sqrt{u} < 12$ and $e < 3$. Also

$$\prod_{i=0}^{m+1}(u + a^2 4^i) \le a^{2m+4} 4^{(m+1)(m+2)/2} \exp\left(\sum_{i \ge 0} \frac{2}{4^i}\right) < 27 a^{2m+4} 4^{(m+1)(m+2)/2}.$$

The numerator is at least $a^2 R$ there. Combining these inequalities proves the lower bound.

Finally,

$$\frac{\mu_{m+1}4^{-(m+1)(m+2)/2}}{\mu_m 4^{-m(m+1)/2}} < \frac{(m+1)^2}{4^{m+2}} \le \frac{1}{16}.$$

Thus the upper bound is at most

$$\frac{t+m+3}{12}\,4^{-t(m+1)}16^{-m} \le \frac{t+m+3}{12}\,4^{-t-m},$$

which tends to zero uniformly as $t+m \to \infty$. □

Theorem 2.2 (Complete compressed classification and raw-error optimum)**.** *Fix m, t and use the preceding notation. Every rational polynomial C of degree at most $m+2$ satisfying*

$$C(1) = 1, \qquad C'(1) = -t, \qquad C(4^{-j}) = 0 \quad (1 \le j \le m)$$

has the unique form

$$C_\kappa(x) = \frac{P_m(x)}{A_m}[1 - b(x-1) + \kappa(x-1)^2], \qquad \kappa \in \mathbb{Q}.$$

This includes every remainder of degree less than $m+3$ obtained from the shared-cutoff polynomial $(x-1)^2(2x-1)P_m(x)$, whenever these normalized constraints hold.

Writing $d = R-1$, its exact gamma-error kernel is

$$F_\kappa(u) = \frac{u^m(u^2 + a^2A_\kappa u + a^4B_\kappa)}{\prod_{i=0}^{m+2}(u + a^2 4^i)},$$

where

$$A_\kappa = 5R + d(b - 3\kappa), \qquad B_\kappa = 4\{R^2 + bRd + \kappa d^2\}.$$

Consequently $F_\kappa(u) \ge 0$ for all $u \ge 0$ if and only if

$$B_\kappa \ge 0, \qquad A_\kappa \ge 0 \text{ or } A_\kappa^2 \le 4B_\kappa. \tag{6}$$

Among all these pointwise nonnegative kernels, the smallest integral is attained uniquely at

$$\kappa_{\min} = -\frac{R(R+bd)}{d^2}.$$

At this parameter $C_{\kappa_{\min}} = C_{m+1,t}$, so the minimum error is exactly $E_{m+1,t}$. This is an optimum for the unnormalized integral; it is not an optimum for the final reduced-denominator error.

Proof. Divisibility by P_m and the two conditions at 1 give the unique quadratic factor. Three shifts of (4) yield the displayed numerator. A monic quadratic $v^2 + Av + B$ is nonnegative on $v \ge 0$ exactly when $B \ge 0$ and either $A \ge 0$ or its real minimum $B - A^2/4$ is nonnegative; this proves (6).

The condition $B_\kappa \ge 0$ forces $\kappa \ge \kappa_{\min}$. At $\kappa_{\min}$, $B_\kappa = 0$ and $A_\kappa > 0$, so this endpoint is admissible. Its polynomial additionally vanishes at 4^{-m-1}, hence the already proved polynomial identity identifies it with $C_{m+1,t}$. The error is the affine function

$$E(\kappa) = E_{m,t} + \kappa J_{m,t}.$$

The preceding proof established $J_{m,t} > 0$ for every m, t, so the minimum over all admissible parameters is attained uniquely at the least one, as claimed. □

A rational double-zero example. For $m = 0$, $t = 4$, and $\kappa = 80/9$, one has

$$C_\kappa(x) = \frac{125 - 196x + 80x^2}{9}, \qquad a = 16,$$

and the exact positive kernel is

$$F_\kappa(u) = \frac{(u - 6144)^2}{(u + 256)(u + 1024)(u + 4096)}.$$

Thus an interior double zero, exact logarithmic cancellation, and global positivity coexist in a three-node formula. No quality claim about its final rational approximation is inferred from that zero.

Theorem 2.3 (An explicit positive certificate with a square-root logarithmic degree bound)**.** *For every rational $p < \gamma$, an exact rational search constructs a globally positive rational kernel of mass exactly $\gamma - p$. When $1/2 \le p < \gamma$, it constructs a dyadic polynomial C with*

$$C(1) = 1, \qquad C'(1) = 0,$$

whose dyadic kernel is positive for all $u > 0$ and whose corrected harmonic target is exactly p. If a separation bound $0 < \varepsilon \le \gamma - p$ is available, any integer $M \ge 32$ satisfying

$$2^{-M^2/2} \le \varepsilon$$

gives the explicit degree bound $\deg C \le M + 1$ and support bound $n \le 2^{M+1}$. In particular the degree bound is $O(\sqrt{\log(1/\varepsilon)})$ as $\varepsilon \to 0$.

For $p = P/Q$ in lowest terms, the complete primitive integrated form is exactly $Q\gamma - P$, regardless of the intermediate interpolation coefficient denominators. The construction does not select a sequence for which these primitive forms tend to zero.

Proof. For $p < 1/2$, use

$$\frac{1}{u+1} + 12(1/2 - p)K_1(u), \qquad K_1(u) = \frac{1}{u+1} - \frac{1}{u+4} + \frac{1}{u+9/4} - \frac{1}{u+9}.$$

Each difference in K_1 is positive, and Binet's integral combined with digamma duplication [4, 5] gives $\int K_1 \, d\mu = 1/12$, proving this assertion directly.

Write $r_m = r_{m,0}$. The preceding theorem gives $r_0 = 1/2$ and $r_m \uparrow \gamma$, with every r_m rational and exactly computable. If $p = 1/2$, take $C = 1$. Otherwise enumerate these exact rationals until the first $m \ge 1$ with $r_m \ge p$. Define

$$\lambda = \frac{p - r_{m-1}}{r_m - r_{m-1}} \in (0, 1], \qquad C = (1 - \lambda)C_{m-1,0} + \lambda C_{m,0}.$$

All coefficients are rational. The constraints and target are exact by linearity, and the kernel is a convex combination of two strictly positive kernels. The search terminates precisely for $p < \gamma$ in the stipulated range $p \ge 1/2$.

For the quantitative bound, the moment formula gives $\mu_m < (2m + 1)!$. Hence

$$E_{m,0} \le (m + 3)(2m + 1)! \, 2^{-m(m+1)}.$$

For $m \ge 32$, the elementary inequalities $2m + 1 \le 2^{m/4}$ and $m + 3 \le 2^{m/2}$ imply

$$(m + 3)(2m + 1)! \le 2^{m/2}(2m + 1)^{2m+1} \le 2^{m^2/2 + 3m/4}.$$

Consequently $E_{m,0} < 2^{-m^2/2}$. If M is as stated, then $r_M > \gamma - \varepsilon \geq p$, so the search terminates with $m \leq M$. The degree and support bounds follow.

The interpolation also has an exact nonnegative integer form. If the two bracketing targets have primitive pairs $(q_0, p_0), (q_1, p_1)$, put $\Delta = q_0 p_1 - q_1 p_0 > 0$. For $p = P/Q$ reduced inside their interval,

$$u = Qp_1 - Pq_1 \geq 0, \qquad v = Pq_0 - Qp_0 \geq 0,$$

and

$$u(q_0, p_0) + v(q_1, p_1) = \Delta(Q, P).$$

Thus a nonnegative integer combination of the two positive primitive kernels gives exactly $\Delta(Q\gamma - P)$, whose full coefficient gcd is Δ. This explicitly tracks the cancellation of all factors of the determinant, including primes already present in the base targets.

Finally choose an integer D that clears the integrated coefficient pair and is divisible by Q. The pair is $(D, DP/Q)$ and its gcd is D/Q, so complete reduction leaves (Q, P). This arithmetic identity does not bound $Q(\gamma - p)$ for targets that have not yet been selected. □

3 The sharp degree scale for positive dyadic certificates

The exact-target constructions allow arbitrary rational coefficients. Their height may therefore grow without restriction. Nevertheless, positivity imposes a quantitative lower bound that depends only on the dyadic support. Together with the canonical construction, this determines the leading asymptotic constant in the smallest possible certificate degree. This is an analytic complexity statement; it does not give primitive smallness.

Lemma 3.1 (A monic polynomial bound on the half-line)**.** *Let $P \in \mathbb{R}[u]$ be monic of degree $d \geq 0$ and nonnegative on $[0, \infty)$. Then*

$$\int_0^1 P(u)\, du \;\geq\; \mathfrak{c}_d := \frac{1}{(d+1)\binom{d}{\lfloor d/2 \rfloor}^2}.$$

The bound is attained among such polynomials.

Proof. First, every polynomial nonnegative on the half-line has a representation

$$P(u) = S(u)^2 + uT(u)^2, \qquad S, T \in \mathbb{R}[u].$$

Here is a direct factorization proof in the monic case. Every positive real root has even multiplicity, so its factor is a square. A zero root contributes powers of $u = 0^2 + u \cdot 1^2$. A negative root contributes $u + a = (\sqrt{a})^2 + u \cdot 1^2$, where $a > 0$. For a pair of nonreal conjugate roots, put $r = \sqrt{b^2 + c^2}$, with $c \neq 0$, and use

$$(u - b)^2 + c^2 = (u - r)^2 + u\big(\sqrt{2(r - b)}\big)^2.$$

Products remain of the stated form because

$$(S^2 + uT^2)(V^2 + uW^2) = (SV - uTW)^2 + u(SW + TV)^2.$$

This proves the representation, including possible repeated roots.

Write $d = 2k + \alpha$, where $\alpha \in \{0, 1\}$. If $\alpha = 0$, the degree and leading coefficient force $\deg S = k$, with leading coefficient 1 or -1, and $\deg T \leq k - 1$. If $\alpha = 1$, they force $\deg T = k$, with leading coefficient 1 or -1, and $\deg S \leq k$. Thus it suffices to bound the squared norm of a monic polynomial of degree k for the weight u^α on $[0, 1]$.

For completeness, the relevant shifted Jacobi polynomial is

$$J_{k,\alpha}(u) = \frac{1}{k!u^\alpha}\frac{d^k}{du^k}\left[u^{k+\alpha}(u-1)^k\right].$$

The expression is a polynomial, with leading coefficient $B = \binom{2k+\alpha}{k}$. Integration by parts k times shows that it is orthogonal, for the weight u^α, to every polynomial of degree less than k. All boundary terms vanish. A second such calculation, followed by the elementary beta integral, gives

$$\int_0^1 u^\alpha J_{k,\alpha}(u)^2\,du = B\int_0^1 u^{k+\alpha}(1-u)^k\,du = \frac{1}{2k+\alpha+1}.$$

Consequently, for every monic V of degree k, orthogonality yields

$$\int_0^1 u^\alpha V(u)^2\,du \geq \int_0^1 u^\alpha\left(\frac{J_{k,\alpha}(u)}{B}\right)^2 du = \frac{1}{(2k+\alpha+1)B^2} = \mathfrak{c}_d.$$

Applying this to S or T, after changing its sign if necessary, proves the bound. Equality is attained by $P = (J_{k,0}/B)^2$ when d is even, and by $P = u(J_{k,1}/B)^2$ when d is odd. Both are nonnegative on the entire half-line. □

Theorem 3.2 (A uniform lower bound at a fixed dyadic cutoff)**.** *Let $a \geq 1$, let $d \geq 0$ be an integer, and let $c_0, \ldots, c_d$ be real numbers satisfying $\sum_{i=0}^d c_i = 1$. Suppose that*

$$\mathcal{K}(u) = \sum_{i=0}^d \frac{c_i}{u + a^2 4^i} \geq 0 \qquad (u \geq 0).$$

For the Binet measure $d\mu(u) = du/(e^{2\pi\sqrt{u}} - 1)$, one has

$$\int_0^\infty \mathcal{K}(u)\,d\mu(u) \geq \frac{\mathfrak{c}_d}{3^7\prod_{i=0}^d(1+a^2 4^i)} \tag{7}$$

$$\geq \frac{1}{3^8(d+1)a^{2d+2}2^{d^2+3d}}. \tag{8}$$

In particular, the bounds hold without any restriction on coefficient height, coefficient signs, or rational denominators.

Proof. Put

$$D(u) = \prod_{i=0}^d (u + a^2 4^i), \qquad P(u) = D(u)\mathcal{K}(u).$$

The numerator P is a polynomial of degree exactly d, and its leading coefficient is $\sum_i c_i = 1$. Since $D > 0$ on the half-line, P is nonnegative there. On $0 < u \leq 1$, the elementary inequalities $2\pi < 7$ and $e < 3$ give

$$\frac{1}{e^{2\pi\sqrt{u}} - 1} > 3^{-7}, \qquad D(u) \leq \prod_{i=0}^d (1 + a^2 4^i).$$

The preceding lemma, applied after restricting the integral to $[0, 1]$, proves (7).

Furthermore,

$$\prod_{i=0}^d (1 + a^2 4^i) = a^{2d+2}2^{d(d+1)}\prod_{i=0}^d (1 + a^{-2}4^{-i}) < 3a^{2d+2}2^{d(d+1)}.$$

Indeed the last product is at most $2\exp(\sum_{i\geq 1} 4^{-i}) = 2e^{1/3} < 3$. Finally $\binom{d}{\lfloor d/2\rfloor} \leq 2^d$, so $\mathfrak{c}_d \geq 1/((d+1)2^{2d})$. These estimates give (8). □

Theorem 3.3 (Sharp asymptotic degree for exact positive dyadic targets)**.** *For a rational number $p \in [1/2, \gamma)$, let $d_*(p)$ be the least degree of a polynomial $C(x) = \sum_i c_i x^i \in \mathbb{Q}[x]$ satisfying*

$$C(1) = 1, \qquad C'(1) = 0, \qquad \mathcal{K}_C(u) := \sum_i \frac{c_i}{u + 4^i} \geq 0 \quad (u \geq 0),$$

and having the exact corrected harmonic target

$$p = \sum_i c_i \left(H_{2^i} - \frac{1}{2^{i+1}} \right).$$

This minimum exists. As rational p approaches γ from below,

$$\frac{d_*(p)}{\sqrt{\log_2(1/(\gamma - p))}} \longrightarrow 1. \tag{9}$$

No assumption about the rationality or irrationality of γ is used. If $p = P/Q$ is reduced, complete integer normalization still leaves exactly $Q\gamma - P$; the degree conclusion is independent of whether this primitive error is small.

Proof. Binet's identity gives

$$H_{2^i} - \frac{1}{2^{i+1}} = \gamma + i \log 2 - \int_0^\infty \frac{d\mu(u)}{u + 4^i}.$$

The two normalization conditions therefore imply

$$\gamma - p = \int_0^\infty \mathcal{K}_C(u)\, d\mu(u) > 0.$$

Put $\delta = \gamma - p$ and $L = \log_2(1/\delta)$. The preceding theorem with $a = 1$ yields, for every admissible degree d,

$$L \leq d^2 + 3d + \log_2(d + 1) + 8 \log_2 3.$$

In particular $d_*(p) \to \infty$ as $\delta \to 0$, and this inequality implies

$$\liminf_{p \uparrow \gamma} \frac{d_*(p)}{\sqrt{L}} \geq 1.$$

We recall the explicit canonical family to prove the matching upper bound. Set

$$P_m(x) = \prod_{j=1}^{m} (4^j x - 1), \quad A_m = P_m(1), \quad s_m = \frac{P_m'(1)}{P_m(1)}, \quad C_m(x) = \frac{P_m(x)}{A_m}[1 - s_m(x - 1)].$$

For $m = 0$, take $P_0 = A_0 = C_0 = 1$ and $s_0 = 0$. These polynomials have $C_m(1) = 1$, $C_m'(1) = 0$, and degree at most $m + 1$. The partial-fraction identity

$$\sum_i \frac{[x^i]P_m(x)}{u + 4^i} = \frac{A_m u^m}{\prod_{i=0}^m (u + 4^i)}$$

follows by comparing the numerator's leading coefficient and its first m Taylor coefficients at zero. Multiplication by x shifts the indices by one; it follows that, with $R_m = 4^{m+1}$,

$$\mathcal{K}_{C_m}(u) = \frac{u^m\{u + R_m + s_m(R_m - 1)\}}{\prod_{i=0}^{m+1}(u + 4^i)} > 0 \qquad (u > 0).$$

Let r_m be its rational corrected target and put $E_m = \gamma - r_m$. In particular $r_0 = 1/2$.

The moment formula

$$\mu_j = \int_0^\infty u^j \, d\mu(u) = \frac{2(2j+1)!\,\zeta(2j+2)}{(2\pi)^{2j+2}}$$

gives $\mu_j < (2j+1)!$ and $\mu_{j+1}/\mu_j < (j+1)^2/4$. The latter inequality follows from $\zeta(2j+4)/\zeta(2j+2) < 1$, $\pi > 3$, and $(2j+3)(2j+2) < 9(j+1)^2$. Also $s_m = m + \sum_{j=1}^m (4^j - 1)^{-1} < m+1$. Bounding each factor in the denominator of the positive kernel below by 4^i now gives

$$E_m \le \mu_m (m+3) 2^{-m(m+1)} < (m+3)(2m+1)!\, 2^{-m(m+1)}.$$

Consequently

$$\log_2 E_m \le -m^2 + O(m \log(m+1)), \qquad E_m \longrightarrow 0.$$

The implied constant is absolute.

For $p > 1/2$, enumerate the rational numbers r_m until the first index $j \ge 1$ with $r_j \ge p$. Such an index exists because $E_m \to 0$. Minimality gives $r_{j-1} < p \le r_j$, and hence

$$\lambda = \frac{p - r_{j-1}}{r_j - r_{j-1}} \in (0,1], \qquad C = (1-\lambda) C_{j-1} + \lambda C_j$$

is an admissible rational polynomial of degree at most $j+1$. Its positivity and exact target follow from linearity and convexity. For $p = 1/2$, simply use $C = 1$. This proves existence of $d_*(p)$ by exact rational comparisons, without an equality oracle for γ.

Fix $\eta > 0$ and take $m = \lceil (1+\eta)\sqrt{L} \rceil$. As $L \to \infty$,

$$\log_2 E_m \le -(1+\eta)^2 L + O(\sqrt{L} \log L) < -L.$$

Thus $E_m < \delta$, so $r_m > p$, and the first crossing index above satisfies $j \le m$. Therefore

$$\limsup_{p \uparrow \gamma} \frac{d_*(p)}{\sqrt{L}} \le 1 + \eta.$$

Letting $\eta \downarrow 0$ proves (9). Finally, an integer D clearing the integrated rational pair gives $(D, DP/Q)$ with gcd D/Q, leaving exactly (Q, P). This verifies the last assertion without a bound on intermediate coefficient heights. □

A quantitative price for positive support compression. Suppose that an original normalized certificate has error $E > 0$, and a target-changing operation replaces it by a globally nonnegative kernel supported at $a2^i$, $0 \le i \le d$, with the same gamma coefficient 1. Assume the complete logarithm condition is preserved, so both masses are log-free gamma errors. If their targets are p and $\widetilde{p}$, then

$$p - \widetilde{p} = (\gamma - \widetilde{p}) - (\gamma - p) \ge \frac{1}{3^8 (d+1) a^{2d+2} 2^{d^2+3d}} - E.$$

In particular, if E is at most half of this cutoff bound, any such positive compression must lower the target by at least half of the bound. This applies to shared-cutoff compression whenever its normalized output kernel remains globally nonnegative. A certificate merely proving that its integrated error is positive does not suffice for this conclusion.

The sharp degree law concerns a fixed base and a specified pole grid. It does not bound arbitrary rational-node representations, support with a varying external base, or signed kernels with only a positive integral. Nor does it control the denominator Q of a target or prove $Q(\gamma - P/Q) \to 0$.

4 Global positive certificates with prescribed local data

The first three results join three requirements that were previously handled separately: control on the whole half-line, exact local vanishing, and an exact rational integral target. They do not provide a bound for the denominator of a well-chosen target.

For $a > 0$ and $u \geq 0$, put

$$h_a(u) = \frac{1}{u + a^2}, \qquad K_a(u) = h_a(u) + h_{a+1/2}(u) - h_{2a}(u) - h_{2a+1}(u), \tag{10}$$

$$k_a(u) = (1+u)K_a(u), \qquad d\mu(u) = \frac{du}{e^{2\pi\sqrt{u}} - 1}, \qquad \ell(f) = \int_0^\infty \frac{f(u)}{1+u}\, d\mu(u). \tag{11}$$

Binet's formula and digamma duplication [4, 5] give

$$\int h_1\, d\mu = \gamma - \frac{1}{2}, \qquad \ell(k_a) = \int K_a\, d\mu = \frac{1}{4a(2a+1)}, \qquad 0 < \ell(1) < \frac{1}{12}. \tag{12}$$

For completeness, Binet gives $I(a) := \int h_a\, d\mu = \log a - \psi(a) - 1/(2a)$. Substituting the duplication identity into $I(a) + I(a+1/2) - I(2a) - I(2a+1)$ yields the rational mass in (12). The total mass of μ is $1/12$, so $0 < \int h_1\, d\mu < 1/12$ follows directly. In particular all the displayed kernel masses are rational when a is rational.

Theorem 4.1 (Density with global and derivative control)**.** *Let $M \geq 0$ be an integer and $L > 0$. The rational linear span of $\{k_a : a \in \mathbb{Q}_{>0}\}$ is dense in the Banach space*

$$X_{M,L} = \{f \in C_0([0,\infty)) : f|_{[0,L]} \in C^M([0,L])\}, \quad \|f\|_X = \|f\|_\infty + \sum_{j=1}^{M} \|f^{(j)}\|_{[0,L]}.$$

No derivative control at infinity is asserted.

Proof. Completeness follows from uniform convergence of the functions and their local derivatives. Suppose a continuous real functional Λ annihilates all rational-node kernels. Hahn–Banach, applied to the embedding of X in $C_0([0,\infty)) \oplus C([0,L])^M$, and the measure representation of the dual give finite signed measures $\nu_0, \ldots, \nu_M$ such that

$$\Lambda(f) = \int f\, d\nu_0 + \sum_{j=1}^{M} \int_{[0,L]} f^{(j)}\, d\nu_j.$$

Apply this integral expression to $g_a(u) = (1+u)/(u+a^2)$ and denote the result by $F(a)$. The expression is meaningful although $g_a \notin C_0$. For $a \geq 1$, $|g_a| \leq 1$ and $g_a(u) \to 0$ at each finite u as $a \to \infty$. Moreover

$$g_a(u) = 1 + \frac{1 - a^2}{u + a^2}$$

shows that all its positive-order derivatives tend uniformly to zero on $[0, L]$. Thus $F(a) \to 0$ by dominated convergence.

Continuity in a extends the annihilation identity from rational nodes to all $a > 0$:

$$F(a) + F(a+1/2) - F(2a) - F(2a+1) = 0.$$

Set $D(a) = F(a) - F(2a)$. Then $D(a+1/2) = -D(a)$, whereas $D(a + j/2) \to 0$. Hence $D = 0$. Iterating $F(a) = F(2a)$ and using the same limit proves $F = 0$.

Write $x = (1+u)^{-1}$. Differentiation with respect to $z = a^2$ at $z = 1$ now gives $\Lambda(x^n) = 0$ for every $n \geq 1$. Differentiation under each measure is justified by uniform bounds in a fixed neighborhood of $z = 1$, including the required local derivatives.

Polynomials in x with zero constant term are dense in X. Here are details for this last approximation step. The change of variable identifies X with functions continuous on $[0,1]$, zero at 0, and C^M on $[b,1]$, where $b=(1+L)^{-1}$; the local derivative norms are equivalent. Extend the Taylor polynomial at b to its left and blend it, in a small interval strictly to the left of b, into a smooth uniform approximation of the remaining continuous function. This gives a global C^M function that is unchanged on $[b,1]$ and arbitrarily close uniformly. A small smooth bump below b restores its value 0 at 0. Approximate its Mth derivative uniformly by a polynomial and integrate M times with the initial derivatives fixed. The resulting polynomial is close in C^M and is still zero at 0. For $M=0$, ordinary polynomial approximation followed by subtraction of the value at 0 suffices. Therefore $\Lambda=0$. This proves real density. Approximation of the finitely many coefficients in each combination by rational numbers proves rational-span density. □

Theorem 4.2 (Exact rational jets and mass with a bounded correction)**.** *Let $T_1,\dots,T_s$ consist of distinct derivative evaluations $f\mapsto f^{(r)}(x)$, with $x\in[0,L]\cap\mathbb{Q}$ and $0\le r\le M$, together with the single functional ℓ. There are positive rational nodes $a_1,\dots,a_s$ for which $C_{ji}=T_j(k_{a_i})$ is an invertible rational matrix. The nodes can be found by exact rational enumeration. For $b\in\mathbb{Q}^s$, define*

$$\mathcal{P}_b(v)=v+\sum_{i=1}^{s}[C^{-1}(b-T(v))]_i k_{a_i}.$$

This map satisfies $T(\mathcal{P}_b(v))=b$ and preserves the rational kernel span. If $T(F)=b$, then

$$\|\mathcal{P}_b(v)-F\|_X\le C_T\|v-F\|_X,\qquad C_T=1+\sum_{i,j}U_i|(C^{-1})_{ij}|V_j,$$

where U_i is any rational upper bound for $\|k_{a_i}\|_X$, and V_j can be taken as 1 for an evaluation and $1/12$ for the mass functional. The upper bound C_T is therefore computable in rational arithmetic.

Proof. The functionals are independent. A smooth positive bump away from the finitely many evaluation points first eliminates the coefficient of ℓ in any relation. Localized polynomial bumps then separate the derivative evaluations. Theorem 4.1 implies full rank on the kernel span. All matrix entries at rational nodes are rational, so enumerating nodes and retaining independent columns terminates. The exact constraint identity follows by multiplication by C. The error bound follows from $b-T(v)=T(F-v)$ and the triangle inequality. Explicit rational norm bounds follow by summing, over the four terms defining k_a,

$$\|g_a\|_\infty\le\max(1,a^{-2}),\qquad \|g_a^{(r)}\|_{[0,L]}\le\frac{r!|1-a^2|}{a^{2r+2}}\quad(r\ge1).$$

Finally $\|\ell\|\le 1/12$ follows from (12). □

Theorem 4.3 (Prescribed zeros and an exact rational target)**.** *Fix finitely many distinct nonnegative rational points x_j and positive integer multiplicities m_j, with m_j even whenever $x_j>0$. Let $d=\sum_j m_j$ and*

$$W(u)=\frac{\prod_j(u-x_j)^{m_j}}{(1+u)^d};$$

use $W=1$ for an empty set. For rational p, the following are equivalent:

(i) $p<\gamma$.

(ii) There exist rational c_i, positive rational a_i, and rational $\eta>0$ such that

$$R(u)=h_1(u)-\sum_i c_iK_{a_i}(u)\ge\frac{\eta W(u)}{1+u}\qquad(u\ge0),\quad\int R\,d\mu=\gamma-p,$$

and the zeros of R on $[0,\infty)$ are exactly the specified points, with exactly the specified multiplicities.

Moreover $R(u) \sim 1/u$ at infinity. If $p = P/Q$ is reduced, complete primitive normalization of its integrated identity gives exactly $Q\gamma - P$, independently of the intermediate coefficient denominators.

Proof. Suppose $\epsilon = \gamma - p > 0$. There exists a smooth function $G > 0$, equal to 1 sufficiently far out, such that $\ell(WG) = \epsilon$. Indeed, let G_{low} be a small positive constant on a long initial interval and 1 in the far tail. Boundedness of W and finiteness of $h_1 d\mu$ show that these masses can tend to 0. On the other hand, $G_{\text{high}} = 1 + C\varphi$, for a nonnegative smooth bump φ away from the zero set, gives arbitrarily large masses. A suitable convex combination realizes ϵ exactly while remaining positive and equal to 1 in the tail.

Put $F_* = 1 - WG$. It belongs to $X_{M,L}$ for $M = \max_j m_j$ and L beyond all the points. At each x_j its value is 1 and its derivatives of orders $1, \dots, m_j - 1$ are zero. Also

$$\ell(F_*) = (\gamma - 1/2) - (\gamma - p) = p - 1/2 \in \mathbb{Q}.$$

For an empty zero set take $M = 0$. Apply Theorems 4.1 and 4.2 to approximate F_* by a rational combination A of the weighted kernels $k_a = (1 + u)K_a$, with all these jets and the mass exact. Near x_j, Taylor's theorem bounds the error by

$$|A(u) - F_*(u)| \le \frac{\|(A - F_*)^{(m_j)}\|}{m_j!} |u - x_j|^{m_j}.$$

The leading coefficient of WG at that point is strictly positive; the parity restriction ensures positivity on both sides of an interior zero. Sufficiently close approximation makes $1 - A$ positive near each point, with precisely that order of vanishing. Away from small disjoint neighborhoods, WG has a positive lower bound, since its limit is 1. Uniform closeness proves positivity there too.

Thus $R = h_1(1 - A)$ has the required zero set and signs. The ratio $(1 - A)/W$ extends positively and continuously across each zero and infinity. Its positive minimum admits a smaller positive rational η, proving the stated global bound. The mass is exact by construction; the tail follows from $K_a(u) = O(u^{-2})$. Conversely, the global bound has positive integral, so it implies $\gamma - p > 0$.

For the arithmetic claim, take a positive integer D clearing the coefficients and both $D/2$ and $D(p - 1/2)$. Then $Q \mid D$, and the integer coefficient pair of the integrated identity is $(D, DP/Q)$. Its gcd is D/Q, giving exactly $Q\gamma - P$. □

Finite verification and effective search. For fixed rational data, enumerate rational kernel combinations and apply the exact projection. The denominators of $R - \eta W/(1+u)$ are positive on the half-line. Clearing them reduces verification to an exact polynomial sign test. Enumerating positive rational η therefore gives a search that terminates exactly when $p < \gamma$. No bound for its degree, height, or running time is asserted. Most importantly, it does not select a sequence for which $Q\gamma - P \to 0$.

Two small certificates can be checked without a search:

$$R_1 = h_1 - \frac{12}{13} K_1 = \frac{u(52u^2 + 325u + 153)}{13(u+1)(u+4)(u+9)(4u+9)}, \tag{13}$$

$$\int R_1 \, d\mu = \gamma - \frac{15}{26};$$

$$R_2 = h_1 + \frac{17}{5640} K_{1/2} - \frac{351}{376} K_1 = \frac{u^2(752u^2 + 4820u + 2943)}{47(u+1)(u+4)(u+9)(4u+1)(4u+9)}, \tag{14}$$

$$\int R_2 \, d\mu = \gamma - \frac{6509}{11280}.$$

Positive numerator coefficients prove the sign globally and give exact orders 1 and 2 at the origin. These are worked finite examples, not evidence for an unproved infinite pattern.

5 Nested rational brackets from the canonical kernels

Define the positive measure and its moments by

$$d\mu(u) = \frac{du}{e^{2\pi\sqrt{u}} - 1}, \qquad \mu_j = \int_0^\infty u^j \, d\mu(u) = \frac{|B_{2j+2}|}{2j+2}.$$

Here B_k denotes a Bernoulli number; the polynomial quantity B_m used later is explicitly defined there and has a different role. Binet's identity [4] gives, for integers $n \geq 1$,

$$H_n - \frac{1}{2n} = \gamma + \log n - \int_0^\infty \frac{d\mu(u)}{u + n^2}. \tag{15}$$

For $m \geq 0$, set

$$P_m(x) = \prod_{j=1}^{m} (4^j x - 1), \quad A_m = P_m(1), \quad s_m = \frac{P_m'(1)}{A_m}, \quad C_m(x) = \frac{P_m(x)}{A_m}[1 - s_m(x-1)].$$

Empty products are 1. Put

$$r_m = \sum_i [x^i] C_m(x) \left(H_{2^i} - \frac{1}{2^{i+1}} \right), \qquad E_m = \gamma - r_m.$$

The first values are $r_0 = 1/2$, $r_1 = 31/54$, and $r_2 = 27269/47250$.

Lemma 5.1 (Positive kernels and a sharpened contraction)**.** *For every $m \geq 0$, with $R = 4^{m+1}$ and $\lambda_m = R + s_m(R-1)$,*

$$E_m = \int_0^\infty \frac{u^m (u + \lambda_m)}{\prod_{i=0}^{m+1} (u + 4^i)} \, d\mu(u) > 0. \tag{16}$$

Define

$$\rho_m = \frac{9(m+1)^2}{16 \cdot 4^{m+1}} \leq \frac{9}{64}.$$

Then

$$0 < \frac{E_{m+1}}{E_m} < \frac{2}{3}\rho_m. \tag{17}$$

In particular r_m increases strictly to γ and $E_{m+1}/E_m \to 0$.

Proof. The polynomial has $C_m(1) = 1$ and $C_m'(1) = 0$, so (15) cancels every logarithm exactly. The elementary rational identity

$$\sum_{i=0}^{m} \frac{[x^i] P_m(x)}{u + 4^i} = \frac{A_m u^m}{\prod_{i=0}^{m} (u + 4^i)}$$

follows by taking a common denominator: its numerator has degree at most m, its first m Taylor coefficients vanish because $P_m(4^{-j}) = 0$ for $1 \leq j \leq m$, and its leading coefficient is $P_m(1) = A_m$. Shifting the indices once evaluates the contribution from xP_m and gives (16).

The exact moment formula also gives the stronger estimate

$$\frac{\mu_{m+1}}{\mu_m} = \frac{(2m+3)(2m+2)}{4\pi^2} \frac{\zeta(2m+4)}{\zeta(2m+2)} < \frac{(m+1)^2}{6}.$$

Indeed $(2m+3)(2m+2) \leq 6(m+1)^2$, $\pi > 3$, and the zeta ratio is strictly less than 1. The kernels F_m in (16) satisfy

$$\frac{F_{m+1}(u)}{F_m(u)} = \frac{u(u + \lambda_{m+1})}{(u + 4R)(u + \lambda_m)} \leq \frac{9u}{4R}.$$

Here $s_{m+1} = s_m + R/(R-1)$ and

$$9\lambda_m - \lambda_{m+1} = \frac{R(R-4)}{R-1} + s_m(5R-8) \geq 0.$$

The function $W_m(u) = (u+\lambda_m)/\prod_{i=0}^{m+1}(u+4^i)$ is strictly decreasing. Its logarithmic derivative is negative because $\lambda_m \geq R$. For the measure $u^m d\mu$, multiplication by a decreasing positive function cannot increase the mean of u. This follows by integrating $(u-v)(W_m(u)-W_m(v)) \leq 0$ against its product measure. Consequently

$$\frac{E_{m+1}}{E_m} \leq \frac{9}{4R}\frac{\mu_{m+1}}{\mu_m} < \frac{3(m+1)^2}{8R} = \frac{2}{3}\rho_m.$$

The ratio is uniformly at most $3/32 < 1$, which proves convergence of E_m to zero. Since $\rho_m \to 0$, the remaining assertions follow. □

Theorem 5.2 (Strictly nested rational upper brackets)**.** *Put $w_m = r_{m+1} - r_m > 0$ and*

$$h_m = r_m + \frac{w_m}{1-\rho_m}, \qquad \delta_m = h_m - \gamma.$$

These quantities satisfy

$$r_m < r_{m+1} < \gamma < h_{m+1} < h_m, \qquad r_m \uparrow \gamma, \quad h_m \downarrow \gamma.$$

More precisely,

$$\frac{\rho_m E_m}{3(1-\rho_m)} < \delta_m < \frac{\rho_m E_m}{1-\rho_m}, \tag{18}$$

and

$$0 < \frac{\delta_{m+1}}{\delta_m} < 2\rho_{m+1}\frac{1-\rho_m}{1-\rho_{m+1}} \leq \frac{9}{32}.$$

Every endpoint r_m, h_m is obtained using rational arithmetic only.

Proof. Since $w_m = E_m - E_{m+1}$,

$$\delta_m = \frac{\rho_m E_m - E_{m+1}}{1-\rho_m}.$$

The sharpened ratio gives the lower bound in (18); positivity of E_{m+1} gives its upper bound. Dividing these bounds at successive indices and using (17) gives the displayed contraction. The sequence ρ_m is nonincreasing, since $\rho_{m+1}/\rho_m = (m+2)^2/(4(m+1)^2) \leq 1$. This proves strict decrease, convergence, and the uniform constant. □

For reference, let $\widehat{p}_m/\widehat{q}_m$ be the first mediant accepted by the Stern–Brocot search in $[r_m, r_{m+1}]$. Unimodularity of the two search parents proves that this is a least-denominator fraction in the interval. The search terminates: any fixed rational in the target interval bounds the increasing parent denominator sum until acceptance. All comparisons use the rational endpoints. Thus

$$0 < \widehat{q}_m\gamma - \widehat{p}_m \leq \frac{\widehat{q}_m w_m}{1-\rho_m}. \tag{19}$$

Also $\widehat{q}_m \to \infty$, because finitely many rational numbers have bounded denominator in $[r_0, 1]$, and each fixed one below γ is eventually below r_m. This proves infinitely many distinct targets, without assuming irrationality. The right side of (19) is an explicit rational upper certificate; its convergence to zero is the unresolved condition.

Theorem 5.3 (Least-denominator selection and an exact rationality dichotomy)**.** *Write $r_m = r_{m,0}$ and define the positive rational numbers*

$$w_m = r_{m+1} - r_m, \qquad \rho_m = \frac{9(m+1)^2}{16 \cdot 4^{m+1}} \le \frac{9}{64}.$$

Choose a reduced fraction $\widehat{p}_m/\widehat{q}_m$ of least positive denominator in the closed rational interval $[r_m, r_{m+1}]$. It can be found by the Stern–Brocot interval algorithm using rational comparisons only; take its first accepted mediant as the definite selection. The denominators satisfy $\widehat{q}_m \to \infty$, without any rationality assumption on γ. Set

$$\mathcal{U}_m = \frac{\widehat{q}_m w_m}{1 - \rho_m}.$$

The associated forms have integer coefficients and satisfy the positive rational certificate

$$0 < \widehat{q}_m \gamma - \widehat{p}_m \le \mathcal{U}_m. \tag{20}$$

For every $j \ge m+1$, the sharper fully rational bound

$$\widehat{q}_m \gamma - \widehat{p}_m \le \widehat{q}_m \left(r_j - \frac{\widehat{p}_m}{\widehat{q}_m} + \frac{w_j}{1 - \rho_j} \right) \tag{21}$$

also holds.

The limit of the completely rational sequence $\mathcal{U}_m$ exists, with the exact dichotomy

$$\lim_{m \to \infty} \mathcal{U}_m = \begin{cases} 0, & \gamma \notin \mathbb{Q}, \\ 1/b, & \gamma = a/b \text{ in lowest terms.} \end{cases} \tag{22}$$

In the rational case the primitive errors $\widehat{q}_m \gamma - \widehat{p}_m$ are eventually exactly $1/b$. Consequently proving $\mathcal{U}_m \to 0$ would prove gamma irrationality; the theorem does not establish which branch in (22) holds.

Proof. Let F_m denote the kernel in (2) at $t = 0$. Put $E_m = \gamma - r_m > 0$ and $\lambda_m = 4^{m+1} + s_m(4^{m+1} - 1)$. The exact kernels give

$$\frac{F_{m+1}(u)}{F_m(u)} = \frac{u(u + \lambda_{m+1})}{(u + 4^{m+2})(u + \lambda_m)}.$$

Writing $R = 4^{m+1}$ and $d = R - 1$, the identity $s_{m+1} = s_m + R/d$ shows

$$\lambda_{m+1} = 4R + s_m(4R - 1) + \frac{R(4R-1)}{d} \le 9\lambda_m.$$

Therefore $F_{m+1}/F_m \le 9u/(4R)$. The function

$$\frac{u + \lambda_m}{\prod_{i=0}^{m+1}(u + 4^i)}$$

is strictly decreasing, since $\lambda_m \ge 4^{m+1}$ and its logarithmic derivative is negative. The same weighted-mean argument as before thus gives

$$\frac{E_{m+1}}{E_m} \le \frac{9}{4R} \frac{\mu_{m+1}}{\mu_m} < \rho_m.$$

In particular $\rho_m \to 0$, and

$$w_m < E_m \le \frac{w_m}{1 - \rho_m}.$$

These bounds prove (20) and (21), because $\widehat{p}_m/\widehat{q}_m \le r_{m+1} < \gamma$.

For completeness, start the Stern–Brocot search with Farey neighbours $0/1$ and $1/0$. Replace the lower neighbour by the mediant when the mediant is below r_m, and replace the upper neighbour when it is above r_{m+1}. Stop when the mediant lies in the target interval. The elementary integral comparison gives $\gamma \le 3/2 - \log 2 < 1$, so the first mediant $1/1$ replaces the upper infinite endpoint. Thereafter both neighbour denominators are positive. The neighbours $a/b < c/d$ always satisfy $bc - ad = 1$. Every reduced fraction strictly between them has denominator at least $b + d$, because its coefficient representation in these two unimodular columns consists of positive integers. The first accepted mediant therefore has least denominator. Termination follows by keeping any one rational in the nonempty target interval: until acceptance the sum of the neighbour denominators increases and is bounded by that rational's denominator. This also proves that the selection uses no gamma oracle.

For any fixed denominator bound, there are only finitely many reduced fractions in the bounded interval $[r_0, \gamma)$ satisfying that bound. Each is eventually smaller than r_m, because $r_m \uparrow \gamma$. Consequently none can be selected for all sufficiently large m, which proves $\widehat{q}_m \to \infty$ and supplies infinitely many distinct targets.

Suppose first that γ is irrational. Fix $0 < \eta < 1/2$. An irrational rotation visits every fixed open arc with bounded gaps. Here is the elementary compactness argument: the pigeonhole principle makes the subgroup generated by γ dense modulo 1; finitely many translates of a fixed open arc cover the circle. A common shift makes their finitely many integer indices nonnegative. Their maximum then bounds a return time from every starting point.

Apply this to the arc $(\eta/4, \eta/2)$. For all large m, there is an integer

$$\frac{\eta}{2E_m} \le q \le \frac{\eta}{E_m} \quad \text{with} \quad \frac{\eta}{4} < \{q\gamma\} < \frac{\eta}{2}.$$

For $p = \lfloor q\gamma \rfloor$ this gives

$$E_m/4 < \gamma - p/q < E_m.$$

Since $E_{m+1}/E_m \to 0$, the fraction lies in $[r_m, r_{m+1}]$ for all large m. Reduction can only lower its denominator, so the minimality of $\widehat{q}_m$ gives $\widehat{q}_m E_m \le \eta$. As η is arbitrary, $\widehat{q}_m E_m \to 0$ and hence $\mathcal{U}_m \to 0$.

Now suppose $\gamma = a/b$ in lowest terms. Every fraction $p/q < \gamma$ has gap at least $1/(bq)$, so $\widehat{q}_m E_m \ge 1/b$. Choose the smallest integer $q \ge 1/(bE_m)$ in the residue class $aq \equiv 1 \pmod b$. Then

$$\frac{1}{bE_m} \le q < \frac{1}{bE_m} + b, \qquad p = \frac{aq-1}{b} \in \mathbb{Z}, \qquad \gcd(p, q) = 1.$$

Its gap satisfies

$$\frac{E_m}{1 + b^2 E_m} < \gamma - p/q \le E_m.$$

For large m this gap exceeds E_{m+1}, so the fraction is in the target interval. Therefore

$$\frac{1}{b} \le \widehat{q}_m E_m \le \frac{1}{b} + bE_m,$$

and $\widehat{q}_m E_m \to 1/b$. Since $w_m/E_m = 1 - E_{m+1}/E_m \to 1$ and $\rho_m \to 0$, this proves the rational branch of (22). Finally each $\widehat{q}_m\gamma - \widehat{p}_m$ is a positive integral multiple of $1/b$ and is at most $\widehat{q}_m E_m \to 1/b$. It must therefore equal $1/b$ for all sufficiently large m. □

6 Logarithmic Farey reduction of the primitive optimization

Let $\alpha = a/b < \beta = c/d < \xi$, where the endpoint fractions are reduced and $b, d > 0$. Set $v_0 = (b, a)$, $v_1 = (d, c)$ and $\Delta = \det(v_0, v_1) = bc - ad > 0$. The linear functional

$$\mathcal{L}_\xi(q, p) = q\xi - p$$

is strictly positive on every nonzero integer vector in the cone spanned by v_0, v_1. The following reduction replaces enumeration of a fundamental parallelogram of size Δ by a logarithmic number of candidates.

Theorem 6.1 (A logarithmic candidate set for the exact primitive minimum)**.** *One can construct, using integer arithmetic only, a set G of primitive vectors (q, p) with $q > 0$ and $\alpha \le p/q \le \beta$ such that*

$$|G| \le \lfloor \log_2 \Delta \rfloor + 2$$

and, simultaneously for every real $\xi > \beta$,

$$\min_{\substack{\alpha \le p/q \le \beta \\ q>0,\ \gcd(p,q)=1}} (q\xi - p) = \min_{(q,p)\in G} (q\xi - p). \tag{23}$$

More generally the same equality holds for every linear functional strictly positive on the nonzero cone. After one extended Euclidean computation for a, b, the construction uses at most $2\lfloor \log_2 \Delta \rfloor + 1$ ceiling-Euclidean updates when $\Delta > 1$. Each collinear run is handled by a single integer division. For $\Delta = 1$, the two endpoint vectors suffice.

The resulting lower envelope is a finite rational piecewise-linear function of ξ. No test for whether ξ equals one of its rational breakpoints is part of the construction.

Proof. The case $\Delta = 1$ follows from the unimodular endpoint basis. Suppose $\Delta > 1$. Choose integers u, v with $bu + av = 1$ and put $r_{\mathrm{raw}} = ud + vc$. A unimodular shear after

$$U = \begin{pmatrix} u & v \\ -a & b \end{pmatrix}$$

sends the two endpoint vectors to $(1, 0)$ and (r, Δ), where $r \equiv r_{\mathrm{raw}} \pmod{\Delta}$ and $1 \le r < \Delta$. The last assertion follows because the upper endpoint is primitive, so $\gcd(r, \Delta) = 1$.

In these transformed coordinates start with

$$z_0 = (1, 0), \quad z_1 = (1, 1), \qquad D_0 = \Delta, \quad D_1 = \Delta - r.$$

While $D_i > 0$, use

$$h_i = \left\lceil \frac{D_{i-1}}{D_i} \right\rceil \ge 2, \qquad z_{i+1} = h_i z_i - z_{i-1}, \qquad D_{i+1} = h_i D_i - D_{i-1}.$$

Then

$$0 \le D_{i+1} < D_i, \qquad \det(z_i, z_{i+1}) = 1, \qquad D_i = \det(z_i, (r, \Delta)).$$

The coordinates are nondecreasing, with increasing second coordinate, so every ray remains in the cone and the terminal vector is exactly (r, Δ): it is a positive parallel primitive vector. The consecutive rays form a unimodular fan. Every integer vector in the cone is therefore a nonnegative integer combination of two adjacent fan vectors. A positive linear functional on such a nonzero combination is at least its value on one of those generators. Thus all fan vectors form an exact candidate set.

When $h_i = 2$, the vector z_i lies on the line segment between z_{i-1} and z_{i+1}. Throughout a run of such updates the vectors form an arithmetic progression, and a linear functional is minimized at one of the run endpoints. Consequently it is enough to retain z_0, the terminal vector, and precisely the vectors z_i for which $h_i \geq 3$.

Every retained interior vertex satisfies $D_i < D_{i-1}/2$. Since the positive determinant remainders decrease, there are at most $\lfloor \log_2 \Delta \rfloor$ such events. This proves the candidate count after applying the inverse unimodular map. All candidate vectors remain primitive, and their first coordinates are positive because they lie in the original cone.

To skip a run with $h_i = 2$, put $g = D_{i-1} - D_i > 0$ and $k = \lfloor D_i/g \rfloor \geq 1$. The next k updates all have quotient 2, and can be replaced by

$$z_{i+k} = z_i + k(z_i - z_{i-1}), \qquad D_{i+k} = D_i - kg.$$

The predecessor is obtained by replacing k by $k-1$. Either the remainder becomes zero or the next quotient is at least 3. There is at most one such run between retained interior vertices, giving the stated update bound. Equality (23) follows from generation and removal of collinear candidates, and rationality of the breakpoints follows by equating two integer affine functions. □

Theorem 6.2 (A terminating relative optimizer without an equality oracle)**.** *Let G be the preceding finite set, and suppose rational brackets $L_j \leq \xi \leq U_j$ are available with $L_j > \beta$ and $U_j - L_j \to 0$. Define*

$$A_j = \min_{(q,p)\in G}(qL_j - p), \qquad B_j = \min_{(q,p)\in G}(qU_j - p).$$

Writing M_ξ for the primitive minimum in (23), one has

$$0 < A_j \leq M_\xi \leq B_j, \qquad B_j - A_j \leq Q_{\max}(U_j - L_j), \quad Q_{\max} = \max_{(q,p)\in G} q.$$

For every rational $\tau > 0$, refinement until $B_j \leq (1+\tau)A_j$ terminates. If $\widehat{v}_j$ is any candidate attaining B_j, then

$$M_\xi \leq \mathcal{L}_\xi(\widehat{v}_j) \leq B_j \leq (1+\tau)M_\xi.$$

At any step all candidates with $qL_j - p > B_j$ may be discarded without losing a true optimum.

For the canonical gamma intervals $[r_m, r_{m+1}]$, the rational brackets

$$L_j = r_j, \qquad U_j = r_j + \frac{r_{j+1} - r_j}{1 - \rho_j}, \qquad j \geq m+2,$$

supply such an optimizer. This gives an explicit infinite family of positive primitive forms within any prescribed fixed relative factor of the true interval optimum. It does not prove that those optima tend to zero.

Proof. For each candidate, replacing ξ by L_j or U_j gives the stated lower and upper bounds. If a vector minimizes the lower expression, its upper expression is at most $A_j + Q_{\max}(U_j - L_j)$; minimizing the upper expression proves the difference estimate. The finite set has a strictly positive minimum M_ξ, so both A_j and B_j tend to M_ξ. Therefore the relative stopping condition holds after finitely many refinements. This argument does not require identifying a unique optimum or deciding whether ξ equals a rational breakpoint. The bound for an upper-minimizing candidate follows directly. A discarded vector has actual value greater than B_j, whereas an undiscarded upper-minimizing vector has actual value at most B_j, so discarding is safe. The previously proved canonical gamma bounds verify all hypotheses of the application. □

Theorem 6.3 (A three-fraction certificate for an exact primitive optimum)**.** *Suppose $a/b < c/d$ are Farey neighbours, so $bc - ad = 1$, and*

$$M = \frac{a+c}{b+d} < \xi < \frac{c}{d}.$$

Then for every reduced fraction $p/q < M$ with $q > 0$,

$$q\xi - p \geq b\xi - a > 0,$$

with equality only for $p/q = a/b$. Consequently, whenever

$$\alpha \leq \frac{a}{b} \leq \beta < M < \xi < \frac{c}{d},$$

the fraction a/b is the unique primitive-error minimizer on $[\alpha, \beta]$, and

$$0 < b\xi - a = \frac{1}{d} - b\left(\frac{c}{d} - \xi\right) < \frac{1}{d}. \tag{24}$$

For gamma, both strict comparisons must be certified: choose rational brackets with $M < L_k \leq \gamma \leq U_j < c/d$. Thus three fractions and rational comparisons certify an exact interval optimum without an equality oracle. In the canonical application below, the next interval's selected mediant satisfies $M < r_{m+2}$, while a computed upper bracket lies below c/d.

Proof. First suppose $p/q \leq a/b$. If $q \geq b$, then $q(\xi - p/q) \geq b(\xi - a/b)$, with equality only at the same primitive fraction. If $q < b$, the strict upper bound $\xi < c/d$ gives

$$(q\xi - p) - (b\xi - a) > \frac{cq - dp - 1}{d} \geq 0,$$

because $cq - dp$ is a positive integer.

If $a/b < p/q < M$, unimodularity gives the positive integers

$$s = cq - dp, \qquad t = bp - aq, \qquad (q, p) = s(b, a) + t(d, c).$$

The inequality $p/q < M$ is equivalent to $s > t$. Hence

$$(q, p) = (s - t)(b, a) + t(b + d, a + c).$$

Both associated linear forms are strictly positive because $a/b < M < \xi$. Their positive integer combination has value strictly greater than $b\xi - a$. This proves the minimum statement. Finally $bc - ad = 1$ gives $c/d - a/b = 1/(bd)$ and therefore (24). □

Proposition 6.4 (A quantitative parent-denominator target)**.** *For the least-denominator fraction of $[r_m, r_{m+1}]$, let its final Stern–Brocot parents be $a_m/b_m < c_m/d_m$. Then*

$$\widehat{q}_m = b_m + d_m, \qquad \widehat{q}_m w_m < \frac{1}{b_m} + \frac{1}{d_m},$$

and consequently

$$\mathcal{U}_m < \frac{b_m^{-1} + d_m^{-1}}{1 - \rho_m}. \tag{25}$$

The lower-parent denominators satisfy $b_m \to \infty$ without assuming gamma irrationality. The upper-parent denominators satisfy the exact alternative:

- *if gamma is irrational, then $d_m \to \infty$;*

- *if gamma is rational, then c_m/d_m is eventually exactly gamma, so d_m is eventually its reduced denominator.*

In particular, an independent proof that d_m is unbounded would suffice for condition 3. This proposition does not supply that proof.

Proof. The selected fraction is the mediant of its final parents, so its denominator is $b_m + d_m$. The target interval lies strictly inside their Farey interval of length $1/(b_m d_m)$, proving the quantitative inequalities.

The selected denominators tend to infinity: otherwise a fixed rational would lie in infinitely many intervals converging strictly from below to gamma, forcing that rational to equal gamma, a contradiction. If the lower denominators had a bounded subsequence, finitely many possible lower parents would give a constant subsubsequence. The upper denominators would then tend to infinity, and the Farey gap would tend to zero, forcing that fixed lower parent to equal gamma. This contradicts its being strictly below every target interval. Thus $b_m \to \infty$.

The same compactness argument shows that a bounded subsequence of d_m forces gamma to be rational. Conversely, if gamma is a reduced rational, the finite Stern–Brocot path reaches it. For all sufficiently late intervals lying strictly below and sufficiently close to it, this rational becomes the right parent during that common initial path. Thereafter mediants with that right parent increase towards gamma. If a mediant overshot the target interval but still lay below gamma, it would replace the right parent. For late intervals this cannot occur before a mediant enters the interval: consecutive such mediants have gaps from gamma of the form $1/[d(b + kd)]$, with successive gap ratio tending to 1, whereas $E_{m+1}/E_m \to 0$. More explicitly, choose the first k with gap at most E_m. Its gap is at least $E_m/(1 + d^2 E_m)$, which exceeds E_{m+1} for all large m. It therefore lies inside the target interval and the search stops with right parent gamma unchanged. This proves eventual constancy. □

Proposition 6.5 (An exact integer test for the final upper Farey parent)**.** *Let $0 < \alpha = p_0/q_0 < \beta = p_1/q_1 < 1$ be reduced rational endpoints. For a proposed reduced upper parent c/d with $\beta < c/d \le 1$, put*

$$k_i = cq_i - dp_i > 0 \quad (i = 0, 1).$$

Choose $b_0 \in \{0, \dots, d-1\}$ with $b_0 c \equiv 1 \pmod{d}$; for $d = 1$ use $b_0 = 0$. Define the integer

$$B_* = b_0 + d \left\lfloor \frac{q_0 - 1 - b_0 k_0}{d k_0} \right\rfloor.$$

Then c/d is exactly the final upper parent in the Stern–Brocot least-denominator search on $[\alpha, \beta]$ if and only if

$$B_* > 0, \qquad (B_* + d) k_1 \le q_1. \tag{26}$$

When these inequalities hold, the lower parent is

$$\frac{A_*}{B_*}, \qquad A_* = \frac{B_* c - 1}{d},$$

and the selected fraction is $(A_* + c)/(B_* + d)$.

For the canonical endpoints, this translates bounded upper-parent denominators into explicit floor and congruence inequalities involving only their harmonic numerator and denominator pairs. In particular,

$$\gcd(q_i, k_i) = \gcd(q_i, d).$$

The existence of many primes dividing q_i and not dividing d merely makes k_i a unit at those primes; it does not by itself contradict (26).

Proof. Set

$$X = \frac{q_0}{k_0} = \frac{1}{d(c/d - \alpha)}, \qquad Y = \frac{q_1}{k_1} = \frac{1}{d(c/d - \beta)}.$$

A Farey lower neighbour A/B of c/d must satisfy $Bc - Ad = 1$, so $Bc \equiv 1 \pmod{d}$. The conditions $A/B < \alpha$ and $(A + c)/(B + d) \geq \alpha$ are respectively

$$B < X, \qquad B + d \geq X.$$

There is exactly one integer in the residue class $b_0 \bmod d$ satisfying these two inequalities: the largest such integer strictly below X. Since $Bk_0 < q_0$ is equivalent to $Bk_0 \leq q_0 - 1$, that integer is exactly B_* in the statement. It must be positive. The remaining condition that the mediant be at most β is $B_* + d \leq Y$, which is precisely the second inequality in (26).

Conversely these inequalities give a Farey pair whose open interval contains $[\alpha, \beta]$ and whose mediant belongs to that target interval. Every fraction in the open Farey interval has denominator at least the sum of the parent denominators, so the mediant is the unique least-denominator fraction in the target interval. A reduced fraction in $(0, 1)$ has a unique Farey parent pair in $[0, 1]$ with positive parent denominators: the right denominator is determined by the inverse congruence modulo the mediant denominator. Thus this pair is exactly the final pair from the Stern–Brocot search. The gcd assertion follows from $\gcd(p_i, q_i) = 1$. □

Why positivity and prime-rich denominators alone do not settle the target. The obstruction can be exhibited exactly for the rational limit $\xi = 1/2$. Choose integers D_m divisible by 4, with $D_{m+1} \geq D_m^2$ and $D_m \to \infty$, and put

$$\ell_m = \frac{1}{2} - \frac{1}{D_m}.$$

The endpoint ℓ_m has reduced denominator exactly D_m, so the D_m can be chosen to include any prescribed increasing lists of primes. The errors have ratio $D_m/D_{m+1} \to 0$. They even have positive integral representations against the same Binet measure: since $\mu_0 = 1/12$,

$$\xi - \ell_m = \int \frac{12}{D_m}\, d\mu > 0.$$

Yet the least-denominator rational in $[\ell_m, \ell_{m+1}]$ is exactly

$$\frac{D_m/4}{D_m/2 + 1},$$

and its primitive error is exactly $1/2$ for every m. Indeed odd denominators require $q \geq D_m/2$ and the smallest eligible odd denominator is $D_m/2 + 1$; even denominators require $q \geq D_m$. The displayed fraction belongs to the interval because $D_{m+1} \geq D_m + 2$. Thus arbitrarily fast raw convergence, global positivity, and prescribed prime factors of the endpoint denominators do not by themselves imply primitive-error decay. This example does not imitate every gamma-specific harmonic coefficient identity.

6.1 Exact finite optima and a rational numerical certificate

In each of the eleven canonical intervals $m = 0, \ldots, 10$, the relative optimizer's surviving set reduces to one vector. The same vector is independently certified by the three-fraction criterion. The following selected rows show the actual optimum, the number of logarithmic-sail candidates, and an upward rational rounding of its positive primitive error.

m	Exact optimal target	Candidates	Strict upper error bound
0	$\frac{4}{7}$	3	$4.05096544 \times 10^{-2}$
1	$\frac{15}{26}$	6	$7.60728744 \times 10^{-3}$
2	$\frac{157}{272}$	15	$2.66085322 \times 10^{-3}$
4	$\frac{33841}{58628}$	49	$1.84706856 \times 10^{-6}$
6	$\frac{9062634}{15700603}$	172	$1.53583719 \times 10^{-9}$
8	$\frac{8788062966751}{15224921118955}$	723	$9.04971062 \times 10^{-14}$
10	$\frac{2323006970406828281}{4024504377931443899}$	2728	$3.81603071 \times 10^{-19}$

For $m = 10$, the certifying triple is

$$\frac{2323006970406828281}{4024504377931443899}, \qquad \frac{3378631047479839813}{5853325287102525206}, \qquad \frac{1055624077073011532}{1828820909171081307}.$$

The first and third fractions are Farey neighbours; the middle is their mediant. The first lies in $[r_{10}, r_{11}]$, while the middle lies strictly above r_{11} and strictly below $r_{12} < \gamma$. The rational upper bracket h_{12} is below the third fraction. Thus the first fraction is the unique primitive-error optimum over the entire interval $[r_{10}, r_{11}]$.

A sharper, independently constructed rational Binet bracket proves

$$\mathcal{F}_{10} = 4024504377931443899\,\gamma - 2323006970406828281,$$
$$\frac{38160307067862345768}{10^{38}} < \mathcal{F}_{10} < \frac{38160307067862345769}{10^{38}}.$$

Here is the exact bracket construction used by the verifier. Put

$$\ell_- = 2\sum_{j=0}^{109} \frac{1}{(2j+1)3^{2j+1}}, \qquad \ell_+ = \ell_- + \frac{2}{221 \cdot 3^{221}}\frac{9}{8}.$$

The positive series for $2\,\mathrm{artanh}(1/3)$ gives $\ell_- < \log 2 < \ell_+$. Define

$$S = \sum_{j=0}^{19} \frac{(-1)^j \mu_j}{64^{2j+2}}, \qquad R = \frac{\mu_{20}}{64^{42}}.$$

Finite division of $1/(64^2 + u)$ with an even number of terms shows

$$S < \int_0^\infty \frac{d\mu(u)}{64^2 + u} < S + R.$$

Consequently the entirely rational endpoints

$$g_- = H_{64} - \frac{1}{128} - 6\ell_+ + S, \qquad g_+ = H_{64} - \frac{1}{128} - 6\ell_- + S + R$$

satisfy $g_- < \gamma < g_+$. Comparing the desired integer form at these two endpoints proves the displayed strict inequalities by integer cross multiplication.

These are finite optimum certificates. They neither prove that the interval optima tend to zero nor assert a computational record for excluding rational denominators of γ.

7 Selection and primitive optimization on shrinking intervals

The next two results clarify the relationship between least-denominator selection and optimization of the actual primitive error. The first is a synthesis of the preceding constructions with standard irrational-rotation and Farey arguments. Its interval hypothesis is weaker than the contraction available for the canonical gamma kernels. The second shows that the two selection rules can nevertheless have arbitrarily different finite errors. Neither result determines whether Euler's constant is rational.

For a real number ξ, define

$$\kappa(\xi) = \begin{cases} 0, & \xi \notin \mathbb{Q}, \\ 1/b, & \xi = a/b \text{ in lowest terms, } b > 0. \end{cases}$$

All rational targets below are reduced, with positive denominators. Their primitive errors are $q\xi - p$; positivity follows from $p/q < \xi$.

Theorem 7.1 (A common limiting dichotomy for three selection rules)**.** *Let $\xi \in \mathbb{R}$ and let*

$$I_m = [\alpha_m, \beta_m], \qquad \alpha_m, \beta_m \in \mathbb{Q}, \qquad \alpha_m < \beta_m < \xi.$$

Put

$$E_m = \xi - \alpha_m, \qquad E_m^+ = \xi - \beta_m.$$

Assume

$$E_m \longrightarrow 0, \qquad \limsup_{m\to\infty} \frac{E_m^+}{E_m} < 1. \tag{27}$$

No monotonicity or nesting of the intervals is required.

Choose $p_m/q_m \in I_m$ of least positive denominator and set

$$S_m = q_m\xi - p_m, \qquad M_m = \min_{\substack{p/q \in I_m \\ q>0,\ \gcd(p,q)=1}} (q\xi - p).$$

The minimum defining M_m is attained and is strictly positive. Then

$$q_m \longrightarrow \infty, \qquad \lim q_m E_m = \lim S_m = \lim M_m = \kappa(\xi). \tag{28}$$

If $\xi = a/b$ is rational in lowest terms, then

$$S_m = M_m = 1/b \qquad \text{for all sufficiently large } m. \tag{29}$$

More generally, suppose a reduced target $P_m/Q_m \in I_m$ satisfies

$$M_m \le T_m := Q_m\xi - P_m \le (1+\tau_m)M_m, \qquad 0 < \tau_m < 1. \tag{30}$$

Then $Q_m \to \infty$ and $T_m \to \kappa(\xi)$. In the rational case, $T_m = 1/b$ eventually. If additionally $\tau_m \to 0$, then

$$\frac{T_m}{M_m} \longrightarrow 1. \tag{31}$$

When the endpoints and narrowing rational brackets for ξ are effectively available, the relative rule can be implemented by Theorem 6.2, for any effectively given rational sequence $0 < \tau_m < 1$. It does not require an equality oracle.

Finally, any positive numbers C_m with $C_m/E_m \to 1$ satisfy

$$q_m C_m \longrightarrow \kappa(\xi). \tag{32}$$

In particular this applies to any asymptotically exact rational upper certificate for E_m.

Proof. For a fixed interval, take any admissible primitive pair (q_*, p_*). Since $E_m^+ > 0$, every admissible pair satisfies

$$q\xi - p = q(\xi - p/q) \geq qE_m^+.$$

Pairs with $q > (q_*\xi - p_*)/E_m^+$ cannot improve the chosen pair. Only finitely many fractions in a bounded interval have a bounded denominator, proving existence and positivity of M_m. The same finiteness observation shows that every sequence of rational targets chosen in I_m has denominator tending to infinity: otherwise a fixed rational would occur along a subsequence converging to ξ strictly from below, forcing that rational to equal ξ.

Choose a fixed number $\theta \in (0, 1)$ such that

$$E_m^+/E_m \leq \theta$$

for all sufficiently large m. We first suppose that ξ is irrational. We use only the bounded-return property of an irrational rotation, not equidistribution. The elementary density of the forward orbit of an irrational rotation implies that, for every nonempty open arc J of $\mathbb{R}/\mathbb{Z}$, the sets $J - j\xi$, $j \geq 0$, cover the circle. Compactness gives a finite subcover. Consequently there is an integer K_J such that, from every starting point, some iterate with index between 0 and K_J lies in J. The density assertion itself follows from the usual pigeonhole approximation of an irrational rotation by arbitrarily small nonzero rotations.

Fix $h > 0$ with $\theta(1 + h) < 1$, and fix $0 < \eta < 1$. Apply the bounded return property to

$$J = (\theta(1 + h)\eta, \eta).$$

For all sufficiently large m, the interval $[\eta/E_m, (1 + h)\eta/E_m]$ is long enough to contain such a return. Thus there is an integer q with

$$\frac{\eta}{E_m} \leq q \leq \frac{(1 + h)\eta}{E_m}, \qquad \theta(1 + h)\eta < \{q\xi\} < \eta.$$

Put $p = \lfloor q\xi \rfloor$. Then

$$E_m^+ \leq \theta E_m < \xi - p/q < E_m.$$

After reduction the fraction still belongs to I_m and its denominator does not increase. Minimality of q_m therefore gives

$$q_m E_m \leq (1 + h)\eta.$$

Since $\eta > 0$ is arbitrary, $q_m E_m \to 0$. The inequalities

$$0 < M_m \leq S_m \leq q_m E_m$$

prove the other two limits in the irrational case.

Now let $\xi = a/b$ be rational in lowest terms. Every positive primitive error is a positive integer multiple of $1/b$, so

$$q_m E_m \geq S_m \geq M_m \geq 1/b.$$

Choose the smallest integer $q \geq 1/(bE_m)$ in the residue class $aq \equiv 1 \pmod{b}$, and put $p = (aq - 1)/b$. For large m, this is a positive denominator and

$$\frac{1}{bE_m} \leq q < \frac{1}{bE_m} + b, \qquad \gcd(p, q) = 1,$$

because $aq - bp = 1$. Its gap satisfies

$$\frac{E_m}{1 + b^2 E_m} < \xi - p/q \leq E_m.$$

For sufficiently large m, the left bound exceeds θE_m and therefore exceeds E_m^+. This fraction belongs to I_m. Hence

$$\frac{1}{b} \le q_m E_m \le \frac{1}{b} + bE_m,$$

which proves $q_m E_m \to 1/b$. Since S_m is a positive integer multiple of $1/b$ and $S_m \le q_m E_m$, eventually $S_m = 1/b$. The inequalities $1/b \le M_m \le S_m$ then give $M_m = 1/b$ as well.

For a relative optimizer, the irrational case follows from $0 < T_m < 2M_m \to 0$. In the rational case, eventually

$$1/b \le T_m \le (1+\tau_m)/b < 2/b.$$

Its membership in $(1/b)\mathbb{Z}$ forces $T_m = 1/b$. If $\tau_m \to 0$, division of (30) by $M_m > 0$ proves (31). The effective assertion uses the finite candidate set of Theorem 6.1 and the terminating refinement rule of Theorem 6.2; for each fixed m, its strict gap $\xi - \beta_m > 0$ lets the available lower rational bracket eventually exceed β_m. Finally (32) follows by multiplying $q_m E_m \to \kappa(\xi)$ by $C_m/E_m \to 1$. □

The exact Farey-parent alternative. Under the hypotheses of Theorem 7.1, the parent criterion also holds for the general intervals, with a short proof that uses the rational stabilization just established. For all sufficiently large m, $q_m > 1$. Let

$$\frac{a_m}{b_m} < \frac{p_m}{q_m} < \frac{c_m}{d_m}, \qquad b_m + d_m = q_m, \quad a_m + c_m = p_m, \quad b_m c_m - a_m d_m = 1,$$

be its canonical Farey parents, whose denominators lie between 1 and $q_m - 1$. They are the final Stern–Brocot parents in the positive-interval application. Since either parent has smaller denominator than q_m, minimality implies

$$a_m/b_m < \alpha_m < \beta_m < c_m/d_m.$$

Their gap is $1/(b_m d_m)$, and consequently

$$q_m(\beta_m - \alpha_m) < \frac{1}{b_m} + \frac{1}{d_m}. \tag{33}$$

One always has $b_m \to \infty$. Indeed, on a subsequence with bounded b_m, one has $d_m = q_m - b_m \to \infty$. The parent gap then tends to zero and forces the lower parents to converge to ξ. A bounded denominator leaves only finitely many fractions near ξ, so some fixed lower parent would equal ξ, contradicting its strict position below $\alpha_m < \xi$. The same argument applied to a bounded subsequence of d_m forces ξ to be rational. Thus irrational ξ implies $d_m \to \infty$.

Conversely, if $\xi = a/b$ is reduced, the theorem gives $aq_m - bp_m = 1$ for all sufficiently large m. Also $q_m > b$ eventually. The right Farey-parent denominator is the unique integer $d \in \{1, \dots, q_m - 1\}$ satisfying $p_m d \equiv -1 \pmod{q_m}$. The displayed determinant identity shows that it is b, with numerator a. Hence

$$\frac{c_m}{d_m} = \xi \quad \text{eventually if } \xi \in \mathbb{Q}. \tag{34}$$

In this class of interval sequences the following statements are therefore equivalent:

$$\begin{aligned} \xi \notin \mathbb{Q} &\iff d_m \to \infty \iff (d_m) \text{ is unbounded} \\ &\iff M_m \to 0 \iff S_m \to 0 \iff q_m E_m \to 0. \end{aligned}$$

These are exact equivalences, not an independent proof of any one of them for $\xi = \gamma$.

Application to the canonical gamma intervals. Take $\alpha_m = r_m$, $\beta_m = r_{m+1}$ and $\xi = \gamma$. Lemma 5.1 supplies the stronger condition $E_m^+/E_m = E_{m+1}/E_m \to 0$, so the theorem applies. The rational upper quantity in Theorem 5.3 is

$$C_m = \frac{r_{m+1} - r_m}{1 - \rho_m}, \qquad \frac{C_m}{E_m} = \frac{1 - E_{m+1}/E_m}{1 - \rho_m} \longrightarrow 1.$$

Thus the original rational upper certificate, the selected primitive errors, and the true interval optima all have the same rationality dichotomy. Choosing, for example, $\tau_m = 1/(m+2)$ in the effective relative optimizer additionally gives an explicit positive primitive sequence whose ratio to the true interval optimum tends to one. Every step terminates using rational comparisons. No uniform running-time bound and no convergence to zero for gamma are inferred.

Theorem 7.2 (Least denominator need not be asymptotically primitive-optimal)**.** *Let $\varphi = (1+\sqrt{5})/2$, $\xi = \varphi^{-1}$, and let $F_0 = 0$, $F_1 = 1$, $F_{k+2} = F_{k+1} + F_k$ be the Fibonacci numbers. Put*

$$r_n = \frac{F_{2n}}{F_{2n+1}} \qquad (n \geq 1).$$

For any integers $N > n \geq 1$, the unique least-denominator fraction in $[r_n, r_N]$ is r_n, whereas the unique primitive-error minimizer there is r_N. Their exact errors are

$$F_{2n+1}\xi - F_{2n} = \varphi^{-(2n+1)}, \qquad F_{2N+1}\xi - F_{2N} = \varphi^{-(2N+1)}. \tag{35}$$

Consequently the quotient of the least-denominator error by the true primitive minimum is exactly

$$\varphi^{2(N-n)}. \tag{36}$$

In particular, with $n_m = 2^m$ and $I_m = [r_{n_m}, r_{n_{m+1}}]$, one has

$$\frac{\xi - r_{n_{m+1}}}{\xi - r_{n_m}} \longrightarrow 0, \qquad \frac{\textit{least-denominator primitive error on } I_m}{\textit{least primitive error on } I_m} = \varphi^{2^{m+1}} \longrightarrow \infty.$$

Thus even the stronger contraction used in the canonical gamma construction does not, by itself, make least-denominator selection asymptotically primitive-optimal.

Proof. The recurrence and $1 - \xi = \xi^2$ give

$$F_{k+1}\xi - F_k = (-1)^k \xi^{k+1}. \tag{37}$$

For example the left side at $k+1$ is $-\xi$ times its value at k, and its value at $k = 0$ is ξ. This proves the identity by induction. Consecutive Fibonacci numbers are coprime, so all fractions in the statement are reduced. The even-index identity puts r_n strictly below ξ, while Cassini's identity gives

$$r_{n+1} - r_n = \frac{1}{F_{2n+1}F_{2n+3}} > 0.$$

Use $r_0 = 0$ for the next parent computation. The fractions

$$\ell_n = \frac{F_{2n-2}}{F_{2n-1}}, \qquad u_n = \frac{F_{2n-1}}{F_{2n}}$$

are Farey neighbours, their mediant is r_n, and

$$\ell_n < r_n \leq r_N < \xi < u_n.$$

The last strict inequality follows from (37) at the odd index $2n-1$. Every reduced fraction strictly between two Farey neighbours has denominator at least the sum of their denominators.

Equality is possible only for their mediant: the coefficient representation in the two unimodular columns consists of positive integers. Here that sum is F_{2n+1}. Since r_n belongs to the target interval, it is the unique least-denominator fraction there.

For the primitive minimum, the fractions

$$\frac{a}{b} = r_N = \frac{F_{2N}}{F_{2N+1}}, \qquad \frac{c}{d} = \frac{F_{2N+1}}{F_{2N+2}}$$

are Farey neighbours with $bc - ad = 1$. Their mediant is r_{N+1}, and

$$r_N < r_{N+1} < \xi < c/d.$$

The three-fraction principle of Theorem 6.3 therefore gives

$$q\xi - p \geq b\xi - a \quad \text{for every reduced } p/q < r_{N+1},$$

with equality only at $p/q = r_N$. The whole interval $[r_n, r_N]$ is within this range, proving its unique primitive optimum.

Equation (37) now proves (35), and their quotient is (36). For the final interval sequence, write $n = n_m$ and $N = n_{m+1}$. Its raw-gap ratio is exactly

$$\frac{\xi - r_N}{\xi - r_n} = \varphi^{-2(N-n)} \frac{F_{2n+1}}{F_{2N+1}} < \varphi^{-2(N-n)} \longrightarrow 0.$$

Since $N - n = 2^m$, the asserted unbounded error ratio follows. □

What the comparison adds. The Fibonacci example does not assert the same behavior for gamma. It establishes that an asymptotic optimality claim cannot be deduced merely from rational interval endpoints, strict approach from below, and a raw-gap ratio tending to zero. Exact primitive optimization can therefore improve finite errors substantially even when a least-denominator selector already has the correct limiting dichotomy. Conversely, the first theorem shows that switching to a relative optimizer does not eliminate the unresolved rationality branch. An independent all-index arithmetic estimate is still required to prove primitive-error decay for Euler's constant.

8 Primitive obstructions for dyadic targets

8.1 Canonical endpoints

Theorem 8.1 (Uniform primitive-cost divergence of the canonical family)**.** *Let $r_{m,t} = p_{m,t}/q_{m,t}$ be reduced with $q_{m,t} > 0$. Except for $(m,t) = (0,0)$, set*

$$N = 2^{t+m+1}, \quad k = t + m + 1, \quad B_{m,t} = tP_m(1) + P_m'(1) > 0, \quad \mathcal{P}_N = \prod_{N/2<\ell\leq N,\ \ell \text{ prime}} \ell.$$

Then

$$\frac{\mathcal{P}_N}{\gcd(\mathcal{P}_N, B_{m,t})} \mid q_{m,t} \tag{38}$$

and

$$q_{m,t}(\gamma - r_{m,t}) \geq \frac{\mathcal{P}_N}{3^{15} k\, 2^{2k(k-1)}}. \tag{39}$$

In particular the primitive errors tend to $+\infty$ along every sequence $m + t \to \infty$, although the raw errors tend to zero and the approximants increase strictly in both coordinates.

Proof. The coefficients of $P_m(1)^2 C_{m,t}$ are integers. Its top coefficient is

$$-4^{m(m+1)/2}\{tP_m(1) + P_m'(1)\}.$$

For an odd prime $\ell \in (N/2, N]$ not dividing $B_{m,t}$, only the top harmonic sum H_N contains a denominator divisible by ℓ, and it contains precisely the term $1/\ell$. All correction terms $1/(2n)$ have power-of-two denominators. Thus the integer-coefficient harmonic sum has ℓ-valuation -1. Division by $P_m(1)^2$ can only decrease that valuation. Every such prime therefore divides the actual reduced denominator, proving (38). Here $N \geq 4$, so all primes in the interval are odd.

Moreover,

$$B_{m,t} = (t + s_m)P_m(1) \leq k\, 4^{m(m+1)/2}.$$

Use this in (38) and multiply by the lower bound in (3). Since $a = 2^t$,

$$q_{m,t}E_{m,t} \geq \frac{\mathcal{P}_N}{3^{15}k\, 2^{2(m+1)(t+m)}} \geq \frac{\mathcal{P}_N}{3^{15}k\, 2^{2k(k-1)}}.$$

The prime number theorem [7] gives $\log \mathcal{P}_N = N/2 + o(N)$, whereas the logarithm of the remaining denominator in this bound is $O((\log N)^2)$. This proves the uniform divergence. All asymptotic estimates here depend only on N; no parameter-dependent short-interval assertion is used.

The conclusion concerns this explicitly specified canonical family. It does not exclude rational choices of κ in the larger compressed class, different moment prescriptions, or a target-changing operation with a separately proved primitive-error bound. □

8.2 Odd-denominator cost of nonperturbative interpolation

We examine arbitrary rational interpolation between consecutive canonical targets. The interpolation coefficient is allowed to have arbitrarily large numerator and denominator. The result excludes every dyadic rational interpolation, including choices whose heights grow without any restriction. For general coefficients it gives an explicit necessary cost in the odd part of their denominator. It does not exclude arbitrary rational interpolation.

Recall the canonical polynomials and rational targets

$$P_m(x) = \prod_{j=1}^{m}(4^j x - 1), \quad A_m = P_m(1), \quad B_m = P_m'(1), \quad s_m = B_m/A_m,$$

$$C_m(x) = \frac{P_m(x)}{A_m}[1 - s_m(x-1)] = \sum_i c_{m,i}x^i, \qquad r_m = \sum_i c_{m,i}f_i, \quad f_i = H_{2^i} - \frac{1}{2^{i+1}}.$$

Here $P_0 = A_0 = C_0 = 1$, $B_0 = s_0 = 0$, and $r_0 = 1/2$. The exact positive-kernel identity is

$$E_m := \gamma - r_m = \int_0^\infty \frac{u^m\{u + 4^{m+1} + s_m(4^{m+1} - 1)\}}{\prod_{i=0}^{m+1}(u + 4^i)} \frac{du}{e^{2\pi\sqrt{u}} - 1} > 0.$$

It also gives the uniform elementary bound

$$E_m \geq 3^{-15}2^{-m(m+1)}. \tag{40}$$

Indeed, restrict the integral to $1 \leq u \leq 2$. Its numerator is at least 4^{m+1}, its denominator is less than $27\, 4^{(m+1)(m+2)/2}$, and the measure density is greater than 3^{-12} on that interval.

Theorem 8.2 (Exact two-adic arithmetic of the canonical targets)**.** *For every integer $m \geq 0$,*

$$v_2(r_m) = -1, \qquad v_2(r_{m+1} - r_m) = 2m + 1.$$

Let $0 < \lambda = u/v \le 1$ be reduced, and write $v = 2^L H$ with H odd. If q is the reduced denominator of $r = r_m + \lambda(r_{m+1} - r_m)$, then

$$v_2(q) = \begin{cases} 1, & 0 \le L \le 2m+1, \\ 0, & L = 2m+2, \\ L - 2m - 1, & L \ge 2m+3. \end{cases}$$

In particular $2^L \le 2^{2m+2} q$ for every such interpolation coefficient.

Proof. Write $P_m(x) = \sum_i a_i x^i$, taking coefficients outside its support to be zero. For $0 \le i \le m$,

$$v_2(a_i) = i(i+1).$$

Indeed, in the elementary symmetric sum of $4, 4^2, \dots, 4^m$, the unique term of least two-adic valuation is the product $4^1 \cdots 4^i$. The number A_m is odd, and for $m \ge 1$ the term a_1 uniquely has least valuation in $B_m = \sum_i i a_i$, so $v_2(B_m) = 2$.

For $m \ge 1$ the coefficients of C_m satisfy

$$c_{m,i} = \frac{(A_m + B_m)a_i - B_m a_{i-1}}{A_m^2}.$$

Consequently $v_2(c_{m,0}) = 0$, $v_2(c_{m,1}) \ge 2$, and

$$v_2(c_{m,i}) = i^2 - i + 2 \quad (2 \le i \le m+1).$$

The unique term of least valuation in H_{2^i} is $1/2^i$, hence $v_2(f_i) = -i - 1$. The $i = 0$ term of r_m therefore has valuation -1, the $i = 1$ term has valuation at least 0, and every $i \ge 2$ term has valuation $(i-1)^2 \ge 1$. This proves $v_2(r_m) = -1$; $m = 0$ is direct.

Put $D_m(x) = P_m(x)(x-1)^2/A_m = \sum_i d_i x^i$ and $Z_m = \sum_i d_i f_i$. From $A_m d_i = a_i - 2a_{i-1} + a_{i-2}$ one gets

$$v_2(d_0) = 0, \quad v_2(d_1) = 1, \quad v_2(d_2) = 0, \qquad v_2(d_i) = (i-2)(i-1) \quad (3 \le i \le m+2).$$

In Z_m, the $i = 2$ term has valuation -3. The terms $i = 0, 1$ have valuation -1, the term $i = 3$ (when present) has valuation -2, and all terms $i \ge 4$ have valuation at least 1. Thus $v_2(Z_m) = -3$.

Set $R = 4^{m+1}$. Direct polynomial subtraction gives

$$C_{m+1} - C_m = -\kappa_m D_m, \qquad \kappa_m = \frac{R\{R + s_m(R-1)\}}{(R-1)^2}.$$

For $m = 0$ this coefficient has valuation 4. For $m \ge 1$, $v_2(s_m) = 2$ and $v_2(R) \ge 4$, so $v_2(\kappa_m) = 2m + 4$. Therefore $r_{m+1} - r_m = -\kappa_m Z_m$ has valuation $2m + 1$.

If $L > 0$, reducedness makes u odd, so the added term has valuation $2m + 1 - L$. If this is greater or less than -1, comparison with $v_2(r_m) = -1$ gives the displayed denominator exponent. At equality, the sum of two rational numbers of valuation -1 is two-adically integral, so the denominator is odd. If $L = 0$, the added term has valuation at least $2m + 1$, and the first branch again applies. □

Theorem 8.3 (A necessary odd-denominator cost for every rational interpolation)**.** *For $m \ge 0$, put $k = m + 1$, $N = 2^{m+2}$, and*

$$\mathcal{P}_N = \prod_{\substack{N/2 < \ell \le N \\ \ell \text{ prime}}} \ell, \quad T_m = A_k B_k > 0, \quad G_m = \frac{\mathcal{P}_N}{\gcd(\mathcal{P}_N, T_m)}.$$

Let $0 < \lambda = u/v \le 1$ be reduced, let H be the odd part of v, and write $r = r_m + \lambda(r_{m+1} - r_m) = p/q$ in lowest terms. For every prime $\ell \mid G_m$, one has the exact formula

$$v_\ell(q) = \max\{0, 1 + v_\ell(v) - v_\ell(u)\}.$$

In particular,

$$G_m \mid qu, \qquad q^2 H \ge \frac{G_m}{2^{2m+2}}. \tag{41}$$

Since

$$T_m < (m+2)2^{2(m+1)(m+2)},$$

the odd denominator must satisfy the completely explicit bound

$$H \ge \frac{\mathcal{P}_N}{(m+2)2^{2(m+1)(m+3)}q^2}. \tag{42}$$

No smallness assumption on λ is used.

Proof. Fix a prime $\ell \mid G_m$. It is odd and does not divide $A_k B_k$. Since $A_m \mid A_k$, the lower target r_m is ℓ-integral: all its harmonic nodes are at most $N/2$, and its coefficient denominators divide A_m^2. The top coefficient of $A_k^2 C_k$ is

$$-4^{k(k+1)/2} B_k.$$

It is an ℓ-unit, and only its harmonic term H_N contains the single nonintegral summand $1/\ell$. The correction $1/(2N)$ is an ℓ-unit. Hence $v_\ell(r_{m+1}) = -1$ and $v_\ell(r_{m+1} - r_m) = -1$.

If $v_\ell(\lambda) < 1$, its product with the increment has negative valuation and dominates the integral base r_m. Otherwise both terms are integral. This proves the exact denominator formula. It implies $\ell \mid qu$ for every $\ell \mid G_m$, hence $G_m \mid qu$. The interval condition gives $u \le v = 2^L H$, and the preceding theorem gives $2^L \le 2^{2m+2} q$. Combining them yields $G_m \le qu \le 2^{2m+2} q^2 H$.

Finally $A_k < 2^{k(k+1)}$ and $B_k = s_k A_k < (k+1)A_k$, because $s_k = k + \sum_{j=1}^{k} (4^j - 1)^{-1} < k + 1$. This proves the bound on T_m and then (42). □

Theorem 8.4 (All dyadic interpolations fail, with unrestricted height)**.** *For the preceding rational interpolation with $0 < \lambda \le 1$,*

$$q(\gamma - r) \ge \frac{\sqrt{\mathcal{P}_N}}{3^{15}\sqrt{(m+2)H}\, 2^{(m+1)(2m+5)}}. \tag{43}$$

Consequently, along every sequence $m \to \infty$ for which the odd part $H = H_m$ of the reduced interpolation denominator satisfies $\log H_m = o(2^{m+2})$, the primitive errors tend to $+\infty$. More quantitatively, with $N_m = 2^{m+2}$,

$$\liminf_{m\to\infty} \frac{\log(q(\gamma - r))}{N_m} \ge \frac{1}{4} - \frac{1}{2} \limsup_{m\to\infty} \frac{\log H_m}{N_m},$$

whenever the right side is finite. Thus divergence also holds when the last limsup is strictly less than $1/2$.

In particular every choice of dyadic rational coefficients $\lambda_m \in [0,1] \cap \mathbb{Z}[1/2]$ fails the primitive-smallness condition, even if their heights grow arbitrarily rapidly. For any rational-interpolation sequence with primitive error tending to zero, it is necessary that

$$\liminf_{m\to\infty} \frac{\log H_m}{2^{m+2}} \ge \frac{1}{2}.$$

The conclusion is a necessary condition, not a construction of weights satisfying it with successful primitive errors.

Proof. The error is the convex combination $(1-\lambda)E_m + \lambda E_{m+1}$, so (40) gives

$$\gamma - r \geq 3^{-15} 2^{-(m+1)(m+2)}.$$

Combine this with $q \geq \sqrt{G_m/(2^{2m+2}H)}$ and $G_m \geq \mathcal{P}_N/T_m$ to obtain (43). The prime number theorem [7] gives $\log \mathcal{P}_N = N/2 + o(N)$, while $m^2 = o(N)$. Taking logarithms proves the quantitative bound and the stated divergence.

For dyadic coefficients $H = 1$. The case $\lambda = 0$, omitted from the divisibility theorem because its reduced numerator is zero, gives the canonical target r_m itself. For $m \geq 1$ this is the preceding interval's interpolation with coefficient 1, so the same proved bound at index $m-1$ already gives divergence. Therefore allowing zero coefficients at any subsequence does not change the all-dyadic conclusion.

Finally, if the primitive errors tend to zero, then they are eventually at most 1. Squaring (43) gives

$$H_m \geq \frac{\mathcal{P}_{N_m}}{3^{30}(m+2)2^{2(m+1)(2m+5)}}$$

at all sufficiently large nonzero-interpolation indices. Zero interpolation cannot occur infinitely often in a successful sequence. Taking logarithms proves the necessary liminf bound. □

Scope. This argument permits completely nonperturbative weights and arbitrary two-adic denominator growth. It does not restrict the height of the odd denominator to be small by assumption and then declare all interpolation impossible: instead it proves how large that odd part must become for a possible successful sequence. A least-denominator target chosen by a Stern–Brocot algorithm may require precisely such enormous odd interpolation denominators, so that selection procedure is not excluded. The exact arithmetic above does not prove its primitive errors tend to zero.

9 Constructive extensions of the positive square map

The results in this section are auxiliary tools for choosing polynomial square forms. They do not prove the irrationality of Euler's constant, and no claim of historical novelty is made. In particular, a degree bound conditional on separation from a rational number is not a proof of that separation.

Put $H_0 = 0$, $H_j = \sum_{h=1}^{j} 1/h$, and

$$a_j = j!(1 + jH_j), \qquad b_j = jj!.$$

The moment normalization is fixed by the exact identity

$$\int_0^\infty t^j (t-1) \log t \, e^{-t} \, dt = \Gamma'(j+2) - \Gamma'(j+1) = j!(1 + jH_j - j\gamma).$$

For polynomials $f, g \in \mathbb{Q}[t]$, define the rational bilinear forms

$$\mathcal{A}(f,g) = \sum_{i,j} f_i g_j a_{i+j}, \qquad \mathcal{B}(f,g) = \sum_{i,j} f_i g_j b_{i+j},$$

and write $A(q) = \mathcal{A}(q,q)$, $B(q) = \mathcal{B}(q,q)$. Thus

$$I(q) := A(q) - \gamma B(q) = \int_0^\infty q(t)^2 (t-1) \log t \, e^{-t} \, dt > 0 \quad (q \neq 0). \tag{44}$$

For integer $q \neq 0$, set $g(q) = \gcd(A(q), B(q)) > 0$ and $E(q) = I(q)/g(q)$. For rational q, first clear coefficient denominators; the resulting primitive pair and $E(q)$ are independent of that choice. For a real parameter θ, write

$$L_\theta(q) = A(q) - \theta B(q) = \int_0^\infty q(t)^2 (t-1)(\log t + \gamma - \theta) e^{-t}\, dt.$$

We use $1/2 < \gamma < 7/12$, which follows directly from the positive Binet integral $\gamma - 1/2 = \int h_1\, d\mu < \int d\mu = 1/12$. For nonnegative integer α, our normalization is

$$L_j^{(\alpha)}(t) = \sum_{h=0}^{j} \frac{(-1)^h}{h!} \binom{j+\alpha}{j-h} t^h.$$

Theorem 9.1 (A quantitative polynomial lift of exponential square tests)**.** *Let $m = 2k \geq 2$, $\lambda > 1$ be rational, and $0 \leq \theta \leq 1$. Define*

$$\alpha = m - 1, \quad \sigma = \frac{\lambda - 1}{2}, \quad \varrho = \frac{\lambda - 1}{\lambda + 1}, \quad c = \left(\frac{2}{\lambda + 1}\right)^m, \qquad f(t) = t^k e^{-\sigma t}.$$

For $N \geq 0$, the polynomial

$$q_N(t) = c\, t^k \sum_{j=0}^{N} \varrho^j L_j^{(\alpha)}(t) \in \mathbb{Q}[t]$$

has degree $N + k$. Put

$$z = \frac{1 + \varrho^2}{2}, \qquad \tau = \frac{\varrho^2}{z}, \qquad \mathcal{S}_\alpha(z) = \left[1 + 64\left(z\frac{d}{dz} + \alpha + 4\right)^3\right](1 - z)^{-\alpha - 1},$$

$$\mathcal{E}_N = 6c^2 \alpha!\, \tau^{N+1} \mathcal{S}_\alpha(z), \qquad \mathcal{F} = 6\left(\frac{(m+2)!}{\lambda^{m+3}} + \frac{(m-1)!}{\lambda^m}\right).$$

These bounds are positive rational numbers and $\mathcal{E}_N \to 0$. The exact exponential test is

$$L_\theta(f) = \frac{m!}{\lambda^{m+2}} \left[(m + 1 - \lambda)(H_m - \log \lambda - \theta) + 1\right], \tag{45}$$

and the entire-ray error satisfies

$$|L_\theta(q_N) - L_\theta(f)| \leq \mathcal{E}_N + 2\sqrt{\mathcal{E}_N \mathcal{F}}. \tag{46}$$

Consequently a certified negative exponential test can be converted into a negative rational polynomial test by a terminating search in N. Clearing the coefficients then gives an integer polynomial with the same sign.

Proof. The Laguerre generating function and orthogonality [6] give

$$e^{-\sigma t} = c \sum_{j \geq 0} \varrho^j L_j^{(\alpha)}(t), \qquad h_j := \int_0^\infty t^\alpha e^{-t} L_j^{(\alpha)}(t)^2\, dt = \frac{(j+\alpha)!}{j!}.$$

These identities follow, respectively, from $(1-u)^{-\alpha-1} \exp(-tu/(1-u))$ and the Rodrigues formula. In the orthonormal Laguerre basis, multiplication by t has diagonal $2j + \alpha + 1$ and off-diagonal $-\sqrt{(j+1)(j+\alpha+1)}$. The sum of absolute values in its jth row is at most $4(j + \alpha + 1)$. Since the matrix is tridiagonal, the row sum for the absolute values of its cube is at most $64(j+\alpha+4)^3$. Symmetry and $2|uv| \leq u^2 + v^2$ therefore imply

$$\int_0^\infty t^\alpha (1 + t^3) e^{-t} \left|\sum_j d_j L_j^{(\alpha)}(t)\right|^2 dt \leq \sum_j h_j [1 + 64(j + \alpha + 4)^3] |d_j|^2.$$

First this holds for finite sums. Applying it to differences of finite tails proves convergence in the displayed weighted norm. The generating function identifies that limit with the exponential, so this also justifies the infinite-tail estimate without a compactness or tail assumption.

For $\theta \in [0,1]$, the signed density factor obeys

$$|(t-1)(\log t + \gamma - \theta)| \le W(t) := 6(t^2 + t^{-1}).$$

Indeed $|\gamma - \theta| < 1$ and $|\log t| \le t + t^{-1}$; bounding separately on $t \le 1$ and $t \ge 1$ proves the stated inequality. With $\|u\|_W^2 = \int W(t)|u(t)|^2 e^{-t}dt$, the preceding Laguerre estimate gives

$$\|q_N - f\|_W^2 \le 6c^2\alpha! \sum_{j>N} \binom{j+\alpha}{\alpha} \varrho^{2j}[1 + 64(j+\alpha+4)^3] \le \mathcal{E}_N.$$

Here $\varrho^{2j} = \tau^j z^j \le \tau^{N+1} z^j$, and the complete sum in z equals $\mathcal{S}_\alpha(z)$. Also $\|f\|_W^2 = \mathcal{F}$. Cauchy–Schwarz applied to $(q_N - f)(q_N + f)$ proves (46).

Finally,

$$\int_0^\infty t^j \log t\, e^{-\lambda t} dt = \frac{j!}{\lambda^{j+1}}(H_j - \gamma - \log\lambda).$$

Substitution and $H_{m+1} = H_m + 1/(m+1)$ give (45). This evaluation also shows explicitly why its only unevaluated logarithm is the elementary $\log\lambda$; the eventual polynomial test has exactly the rational coefficients $A(q_N) - \theta B(q_N)$ when θ is rational. □

Theorem 9.2 (An effective degree bound for exclusion and representation)**.** *Let $0 < \varepsilon \le 1/3$ and put*

$$m = 2\left\lceil \frac{48}{\varepsilon^2} \right\rceil, \qquad N = 8m^2, \qquad D = N + \frac{m}{2} < 77000\varepsilon^{-4}.$$

For every real $\theta \in [0,1]$ with $|\theta - \gamma| \ge \varepsilon$, there is a rational polynomial q of degree at most D such that $L_\theta(q) < 0$. For rational θ this polynomial has an explicit construction using only rational arithmetic and certified bounds for $\log m$.

Let

$$S_d = \{X \in \mathbb{R} : [a_{i+j} - Xb_{i+j}]_{0\le i,j\le d} \text{ is positive semidefinite}\}.$$

Then

$$S_D \subset (\gamma - \varepsilon, \gamma + \varepsilon). \tag{47}$$

For every rational r with $|r - \gamma| \ge \varepsilon$, there is moreover an integer polynomial $q \ne 0$ of degree at most D such that $B(q) \ne 0$ and $A(q)/B(q) = r$. A search restricted to this degree therefore terminates. No coefficient-height bound is asserted.

Proof. Choose rational $\widetilde{h}$ with

$$|\widetilde{h} - (H_m - \log m - \theta)| \le 1/m, \qquad s = \widetilde{h}/2, \qquad \lambda = m(1+s).$$

For rational θ this uses no digits or assumption about γ. For example, for $u = (m-1)/(m+1)$ the partial sums of $2\sum_{j\ge0} u^{2j+1}/(2j+1)$ approximate $\log m$ with remainder at most $2u^{2J+3}/((2J+3)(1-u^2))$ after index J. The elementary estimate

$$0 < H_m - \log m - \gamma < 1/m$$

follows by summing $0 < \log(1+1/j) - 1/(j+1) < 1/(j(j+1))$ for $j \ge m$. Consequently $|s| \le 1/3$ and $|s| \ge \varepsilon/4$. In particular $2m/3 \le \lambda \le 4m/3$ and $\lambda > 1$.

Write $h = H_m - \log m - \theta$ and $e = h - 2s$, so $|e| \le 1/m$. On $[-1/3, 1/3]$ the function $g(s) = 2s - \log(1+s)$ satisfies

$$sg(s) \ge \tfrac{1}{2}s^2, \qquad |g(s)| \le \tfrac{5}{4}|s|,$$

as follows from $1/2 \le g'(s) \le 5/4$. The bracket in (45) is thus

$$1 + (1 - ms)(g(s) + e) \le 1 + \tfrac{9}{4}|s| + 1/m - \tfrac{m}{2}s^2 < 2 - \tfrac{m}{2}s^2 \le -1.$$

It follows that $L_\theta(f) \le -\eta$, where $\eta = m!/\lambda^{m+2} > 0$ is rational.

We record estimates establishing a fully explicit truncation order. With the notation of Theorem 9.1, $m \ge 96$ and elementary binomial generating functions give

$$\mathcal{S}_{m-1}(z) \le 257m^3(1-z)^{-m-3}.$$

For clarity, use $(j+m+3)^3 \le 2(j+m)^3$ for $m \ge 32$ and

$$(j+m)^3\binom{j+m-1}{m-1} \le m(m+1)(m+2)\binom{j+m+2}{m+2},$$

then $m(m+1)(m+2) \le 2m^3$. Since

$$1 - z = \frac{2\lambda}{(\lambda+1)^2}, \qquad \tau = 1 - \frac{2\lambda}{\lambda^2+1},$$

one obtains

$$\frac{\mathcal{E}_N}{\eta} < 60000m^7 2^m \exp\left(-\frac{3(N+1)}{4m}\right), \qquad \frac{\mathcal{F}}{\eta} \le 24m.$$

For the first inequality, use

$$6c^2(m-1)!\mathcal{S}_{m-1}(z) < 30000m^6(m-1)!(2/\lambda)^m$$

and $\tau^{N+1} \le \exp(-3(N+1)/(4m))$. For $N = 8m^2$,

$$\mathcal{E}_N < \frac{\eta}{1536m}.$$

Indeed the logarithm of $92160000m^8 2^m e^{-6m}$ is negative for $m \ge 96$; the inequalities

$$\log(92160000) < 27, \qquad 8\log m \le 2m, \qquad m\log 2 < m$$

suffice. Thus

$$\mathcal{E}_N + 2\sqrt{\mathcal{E}_N\mathcal{F}} < \eta/2,$$

and the constructed q_N has $L_\theta(q_N) < -\eta/2$.

Each S_d is a convex interval, contains γ, and the intervals are nested as d grows. The tests at $\theta = 0, 1$ exclude both endpoints, since their distances from γ exceed $1/3$. Therefore $S_D \subset (0,1)$; the other tests give (47). The degree estimate follows from $m \le 98\varepsilon^{-2}$ and $8 \cdot 98^2 + 49 < 77000$.

For rational $r \in [0,1]$ at the stated distance, the rational quadratic form L_r on polynomials of degree at most D has a negative vector and the positive vector 1. For $r < 0$, use the test negative at 0: its B is negative by (44), so it remains negative at every $r < 0$. Likewise the test negative at 1 remains negative for $r > 1$. Thus the form is indefinite in every case. Its dimension is at least five. If singular, a nonzero rational kernel vector is isotropic. Otherwise Meyer's theorem for indefinite rational quadratic forms in at least five variables supplies a rational isotropic vector. Clearing denominators gives $q \in \mathbb{Z}[t] \setminus \{0\}$. Equation (44) forces $B(q) \ne 0$, and hence $A(q)/B(q) = r$. Enumeration of integer coefficient vectors of the bounded degree is a terminating exact algorithm; no bound for its running time or the coefficient height is implied. □

A quantitative enclosure and its scope. For every integer $d \geq 6237000$, the same argument yields the explicit enclosure

$$S_d \subset \left(\gamma - (77000/d)^{1/4}, \gamma + (77000/d)^{1/4}\right).$$

The condition $|r - \gamma| \geq \varepsilon$ remains essential in the representation theorem. Applying it to an arbitrary rational r without first establishing that condition would assume the desired exclusion.

10 Exact arithmetic in Laguerre coordinates

This section makes the coefficient arithmetic of the positive square map explicit in the ordinary Laguerre basis. It provides an optimal universal normalization, a local prime-residue rule, and a full primitive-cost classification for fixed polynomial coefficient schedules. It does not exclude arbitrary growing sequences of coefficient vectors.

Let

$$L_j(t) = \sum_{h=0}^{j} \frac{(-1)^h}{h!}\binom{j}{h} t^h, \qquad H_0 = 0, \quad H_j = \sum_{h=1}^{j} \frac{1}{h}.$$

For a rational polynomial Q, define the rational numbers $A(Q), B(Q)$ by

$$A(Q) - \gamma B(Q) = I(Q) := \int_0^\infty Q(t)^2(t-1)\log t\, e^{-t}\, dt, \qquad B(Q) = \int_0^\infty Q(t)^2(t-1)e^{-t}\, dt.$$

Their rationality also follows from the matrices below. For every nonzero Q, $I(Q) > 0$. Clear denominators in $(A(Q), B(Q))$ by any positive integer D and let $g = \gcd(DA(Q), DB(Q))$. The primitive error is $E(Q) = DI(Q)/g$, independent of the choice of D or nonzero rational scaling of Q. If $B(Q) \neq 0$ and q is the reduced denominator of $A(Q)/B(Q)$, then

$$E(Q) = q\,\frac{I(Q)}{|B(Q)|}. \tag{48}$$

If $B(Q) = 0$, then $E(Q) = 1$.

Theorem 10.1 (Optimal half-lcm normalization in Laguerre coordinates)**.** *Let $Q_c = \sum_{j=0}^{n} c_j L_j$ with $c_j \in \mathbb{Z}$. Then $B(Q_c)$ is an integer. For $n \geq 2$, the smallest positive integer D_n such that $D_n A(Q_c) \in \mathbb{Z}$ for every $c \in \mathbb{Z}^{n+1}$ is*

$$D_n = \frac{\operatorname{lcm}(1, \ldots, n)}{2}.$$

For $n = 0, 1$ it is $D_n = 1$. The exact symmetric Gram matrices are

$$B_{ii} = 2i, \qquad B_{i,i+1} = -(i+1), \qquad B_{ij} = 0 \quad (|i-j| \geq 2), \tag{49}$$

$$A_{ii} = 2iH_i + 2i + 1, \qquad A_{i,i+1} = -(i+1)H_i - \frac{3i}{2} - 1, \tag{50}$$

$$A_{ij} = \frac{i + j + (j-i)^2}{(j-i)((j-i)^2 - 1)} \qquad (j - i \geq 2). \tag{51}$$

Thus $(D_n A(Q_c), D_n B(Q_c))$ is an exact integer coefficient pair, without requiring factorial clearing of the polynomial coefficients. Its gcd must still be removed to obtain its primitive pair.

Proof. The Laguerre recurrence is

$$tL_i = (2i+1)L_i - iL_{i-1} - (i+1)L_{i+1},$$

and the L_i are orthonormal for $e^{-t}dt$. This gives (49). The logarithmic Gram matrix is

$$\int_0^\infty \log t\, L_i(t)L_j(t)e^{-t}\,dt = \begin{cases} H_i - \gamma, & i = j, \\ -1/|i-j|, & i \neq j. \end{cases}$$

One proof differentiates generalized Laguerre orthogonality at parameter zero. The norm derivative is $H_i - \gamma$, and

$$\partial_\alpha L_i^{(\alpha)}\Big|_{\alpha=0} = \sum_{j<i} \frac{L_j}{i-j}.$$

The last identity follows by differentiating the Laguerre generating function; see [6]. All expressions are finite polynomial sums and the differentiated integrals have an integrable majorant. Multiplying this logarithmic matrix by the recurrence for $t-1$, including the L_{n+1} term when necessary, gives (50)–(51).

Write $d_n = \operatorname{lcm}(1,\dots,n)$. A quadratic form is integer-valued on all integer vectors exactly when its diagonal coefficients and twice its off-diagonal coefficients are integers. For $k = j - i \geq 2$, the useful partial-fraction identity is

$$2A_{ij} = \frac{2(i+1)}{k-1} - \frac{4i}{k} + \frac{2i}{k+1}.$$

If $i \geq 1$, all displayed denominators are at most n. If $i = 0$, only $2/(k-1)$ remains. Together with (50), this shows that $d_n/2$ clears the quadratic values.

For sharpness, testing the vectors e_i and $e_i + e_j$ forces every denominator in A_{ii} and $2A_{ij}$. Row zero supplies $2/(k-1)$ for $2 \leq k \leq n$, so for $n \geq 3$ it forces $d_{n-1}/2$. Induct on n, starting at $n = 2$. A new factor is needed only when n is a prime power. For $n \geq 3$,

$$2A_{1n} = \frac{2[n(n-1)+2]}{(n-1)n(n-2)}.$$

If $n = p^a$ for an odd prime p, this has p-adic valuation $-a$. If $n = 2^a$ with $a \geq 2$, its valuation at 2 is $1 - a$. These are exactly the new prime-power factors of $d_n/2$, proving optimality. □

For finite vectors c, d, pad both by zero and write

$$\mathcal{B}(c,d) = \sum_{r\geq 0} 2r\, c_r d_r - \sum_{r\geq 0} (r+1)(c_r d_{r+1} + c_{r+1} d_r).$$

Thus $B(Q_c) = \mathcal{B}(c,c)$.

Theorem 10.2 (Exact residue at a prime in the upper half)**.** *Let p be an odd prime with $n/2 < p \leq n$, and let $s = n - p$. For an integer coefficient vector $c = (c_0,\dots,c_n)$ put*

$$d_r = c_{p+r}\ (0 \leq r \leq s), \qquad h_r = c_r\ (0 \leq r \leq s+1),$$

and pad these vectors by zero. Then $pA(Q_c)$ is p-integral and

$$pA(Q_c) \equiv \mathcal{B}(d,d) - 2\mathcal{B}(h,d) \pmod{p}. \tag{52}$$

If this residue is nonzero and $B(Q_c) \neq 0$, the prime p divides the actual reduced denominator of $A(Q_c)/B(Q_c)$.

Proof. The normalization theorem shows that $v_p(A(Q_c)) \geq -1$. In the diagonal terms, the residue of pA_{ii} is $2i$ when $i \geq p$, and zero otherwise. In the adjacent cross terms, the residue of $2pA_{i,i+1}$ is $-2(i+1)$ when $i \geq p$, and zero otherwise. These give $\mathcal{B}(d,d)$.

For a gap $k = j - i \geq 2$, the only potentially nonzero residues of $2pA_{ij}$ occur at $k = p-1, p, p+1$. They are, respectively,

$$2i, \qquad -4i, \qquad 2(i+1).$$

The possible exceptional gap $2p-1$ has $i = 0$ and residue zero. Collecting the three contributing diagonals gives

$$2\sum_{r=0}^{s}(r+1)h_{r+1}d_r - 4\sum_{r=0}^{s} rh_r d_r + 2\sum_{r=1}^{s} rh_{r-1}d_r = -2\mathcal{B}(h,d).$$

At $n = 2p-1$ the apparent overlapping endpoint has coefficient $s+1 = p$ and vanishes modulo p, so the same formula applies.

If the residue is nonzero, $v_p(A(Q_c)) = -1$. Since $B(Q_c)$ is an integer, $v_p(A(Q_c)/B(Q_c)) = -1 - v_p(B(Q_c)) < 0$. This proves survival in the reduced denominator, not merely in a chosen clearing factor. □

Theorem 10.3 (Complete primitive-cost classification of fixed polynomial schedules)**.** *Fix a nonzero polynomial $P(X,Y) \in \mathbb{Q}[X,Y]$ and put*

$$Q_N(t) = \sum_{k=0}^{N} P(N,k)L_k(t).$$

These polynomials are nonzero for all sufficiently large N. The quantity $B_N = B(Q_N)$ is a rational polynomial in N. Exactly one of the following holds:

1. *B_N is identically zero. Then $E(Q_N) = 1$ for all sufficiently large N.*
2. *B_N is not identically zero. Then it is eventually nonzero and*

$$\liminf_{N\to\infty} \frac{\log E(Q_N)}{N} \geq \frac{1}{2}.$$

In particular, no fixed polynomial coefficient schedule of this form has primitive error tending to zero.

Proof. Clear the fixed coefficient denominators of P once. This scales each Q_N by the same nonzero rational factor and leaves its primitive error unchanged, so assume $P \in \mathbb{Z}[X,Y]$.

Write $c_k = P(N,k)$ and define the integer-valued rational polynomial

$$F_P(N) = 2(N+1)P(N,N)P(N,N+1) - B_N. \tag{53}$$

Both B_N and $F_P(N)$ are polynomials because sums of polynomial functions over $k = 0, \ldots, N$ are polynomials in N. We first prove that F_P is nonzero; it is essential not to infer this merely from nonconstancy in the second variable.

Let d be the total degree of P. Its highest homogeneous part gives a nonzero real polynomial f such that, uniformly for $x \in [0,1]$,

$$P(N, Nx) = N^d f(x) + O_P(N^{d-1}).$$

The exact discrete energy identity is

$$B_N = (N+1)c_N^2 - \sum_{k=0}^{N} c_k^2 + \sum_{k=0}^{N-1}(k+1)(c_k - c_{k+1})^2. \tag{54}$$

The last sum is $O_P(N^{2d})$, while the middle sum is a polynomial Riemann sum. Hence

$$B_N = N^{2d+1}\left[f(1)^2 - \int_0^1 f(x)^2\,dx\right] + O_P(N^{2d}),$$

and therefore

$$F_P(N) = N^{2d+1}\left[f(1)^2 + \int_0^1 f(x)^2\,dx\right] + O_P(N^{2d}). \tag{55}$$

The coefficient in brackets is strictly positive. Thus F_P has degree exactly $2d+1$ and is nonzero for all sufficiently large N.

Put $r = \deg_Y P$. Let p be a prime with

$$N/2 < p \le N, \qquad p > 2r+2,$$

and let $s = N - p$. In (52), $P(N, p+j) \equiv P(N, j) \pmod p$. Thus the residue becomes

$$pA(Q_N) \equiv 2(s+1)P(N,s)P(N,s+1) - B_s(P(N,\cdot)) \pmod p.$$

For fixed N the right side is a rational polynomial in s. Its coefficient denominators have no prime factors above $2r+2$: expand the polynomial summands of degree at most $2r+1$ in the Newton binomial basis and sum them. Since $s \equiv N \pmod p$, we may replace s by N and conclude

$$pA(Q_N) \equiv F_P(N) \pmod p. \tag{56}$$

When $B_N \neq 0$, every prime in $(N/2, N]$ not dividing $F_P(N)$ therefore survives in the reduced denominator q_N of $A(Q_N)/B_N$. For all sufficiently large N the fixed lower restriction $p > 2r+2$ is automatic. Since the product of the distinct excluded primes divides the nonzero integer $F_P(N)$,

$$\log q_N \ge \vartheta(N) - \vartheta(N/2) - \log|F_P(N)| = \left(\frac{1}{2} + o(1)\right) N.$$

Here $\vartheta(x) = \sum_{p \le x} \log p$ and $\vartheta(x) \sim x$ is the prime number theorem [7].

We must also control the real error; denominator growth alone would not suffice. If Q has degree j and highest Laguerre coefficient c_j, then

$$I(Q) \ge e^{-2} \log 2\,(j+1)c_j^2. \tag{57}$$

Indeed, restrict the positive integral to $t \ge 2$ and use $(t-1)\log t \ge (t-2)\log 2$. After $t = x+2$, the leading monomial coefficient is $(-1)^j c_j/j!$. The monic degree-j orthogonal polynomial for $xe^{-x}dx$ has squared norm $j!(j+1)!$, so the orthogonal decomposition of $Q(x+2)$ gives

$$I(Q) \ge e^{-2} \log 2 \left(\frac{c_j}{j!}\right)^2 j!(j+1)!,$$

which is (57). The norm identity follows directly from the generalized Laguerre Rodrigues formula by j integrations by parts.

For all sufficiently large N, $P(N, Y)$ is a nonzero polynomial of degree at most r. It cannot vanish at the $r+1$ distinct integers $N, N-1, \ldots, N-r$. Consequently $\deg Q_N \ge N - r$, and its highest Laguerre coefficient is a nonzero integer. Thus $I(Q_N)$ has a positive linear lower bound. Since the nonzero polynomial B_N has polynomial size, $I(Q_N)/|B_N|$ is bounded below by an inverse power of N. Equation (48) now proves the exponential primitive-error lower growth.

If B_N is identically zero, positivity gives $A(Q_N) > 0$ whenever $Q_N \neq 0$. Primitive normalization of the rational pair $(A(Q_N), 0)$ then gives error exactly one. □

Scope and an instructive zero-coefficient case. Nonconstant coefficient profiles can satisfy $B_N \equiv 0$. For example, $P(N,k) = 3k-(2N+1)$ has this property; the first branch of the theorem includes it. The proof of (55) remains valid in this case and does not assume that B_N is nonzero.

Let

$$\mathcal{D}_t = t\frac{d^2}{dt^2} + (1-t)\frac{d}{dt}, \qquad \mathcal{D}_t L_k = -kL_k.$$

The theorem covers every fixed polynomial filter $P(N,-\mathcal{D}_t)$ applied to the zero-location kernel $\sum_{k=0}^{N} L_k(t)$. It therefore covers polynomial tapers of any fixed order and every fixed polynomial dependence on N. Coefficients such as $L_k(x_N)$ at nonzero x_N, filters whose degree grows with N, and arbitrary coefficient sequences are outside its scope.

11 Shared-cutoff support compression

Theorem 11.1 (Compression at a shared cutoff and its primitive cost)**.** *For $m \geq 0$ define*

$$T_m(x) = (x-1)^2(2x-1)\prod_{k=1}^{m}(4^k x - 1), \qquad d = m+3, \qquad s = 1+m(m+1).$$

For every $C \in \mathbb{Z}[x]$, division gives unique $S, R \in \mathbb{Z}[1/2][x]$ with

$$C = T_m S + R, \qquad \deg R < d.$$

If $\deg C = L \geq d$, all coefficient denominators of S and R divide $2^{s(L-d+1)}$. Moreover,

$$\begin{aligned} &R(1) = C(1), && R'(1) = C'(1), && R(1/2) = C(1/2), \\ &R(4^{-k}) = C(4^{-k}) && && (1 \leq k \leq m). \end{aligned} \tag{58}$$

Let $a = 2^t$, where $t \geq 0$ is an integer, and assume

$$Q = C(1) \neq 0, \qquad C'(1) + tQ = 0. \tag{59}$$

For $C(x) = \sum c_i x^i$ and $R(x) = \sum r_i x^i$, define the corrected harmonic targets

$$r_C = \frac{1}{Q}\sum_i c_i\left(H_{a2^i} - \frac{1}{2a2^i}\right), \qquad r_R = \frac{1}{Q}\sum_i r_i\left(H_{a2^i} - \frac{1}{2a2^i}\right). \tag{60}$$

Both give log-free gamma forms. Their target difference is generally nonzero, but it satisfies the exact identity and bound

$$\begin{aligned} \Delta := r_C - r_R &= \frac{1}{Q}\sum_i [x^i](T_m S)H_{a2^i}, \\ |\Delta| \leq \Delta_{\max} &:= \frac{\mu_m a^{-2m-2}}{|Q|} T_m^{\mathrm{abs}}(x_0) S^{\mathrm{abs}}(x_0), \qquad x_0 = 4^{-m-1}, \end{aligned} \tag{61}$$

where $P^{\mathrm{abs}}(x) = \sum_i |[x^i]P| x^i$.

Let E be the least nonnegative integer with $2^E R \in \mathbb{Z}[x]$, and put

$$N_0 = a2^{d-1}, \qquad D_0 = 2\,\mathrm{lcm}(1,\ldots,N_0), \qquad Z = 2^E D_0 \sum_i r_i\left(H_{a2^i} - \frac{1}{2a2^i}\right) \in \mathbb{Z}.$$

The complete reduced denominator of r_R is exactly

$$q_R = \frac{|Q|2^E D_0}{\gcd(|Z|, |Q|2^E D_0)}. \tag{62}$$

In particular,

$$\operatorname{odd}(q_R) \mid \operatorname{odd}(Q)\operatorname{odd}\bigl(\operatorname{lcm}(1,\dots,N_0)\bigr).$$

Every odd prime above N_0 not dividing Q is therefore absent after compression, even if it appeared in the original target r_C.

If independently certified rational bounds satisfy

$$0 < \ell \le \gamma - r_C \le u, \qquad \Delta_{\max} < \ell,$$

then, with $P_R = q_R r_R \in \mathbb{Z}$,

$$0 < \ell - \Delta_{\max} \le \gamma - r_R \le u + \Delta_{\max}, \qquad 0 < q_R\gamma - P_R \le q_R(u + \Delta_{\max}). \tag{63}$$

This is a checkable transfer of a positive error certificate. It is not an assertion that the last upper bound tends to zero.

Proof. The leading coefficient of T_m is

$$2\prod_{k=1}^{m} 4^k = 2^{1+m(m+1)} = 2^s.$$

Long division therefore introduces only powers of 2 in denominators. Each of the at most $L-d+1$ steps increases the necessary power by at most s, giving the stated bound. The double root at 1 and the other listed roots give (58). In particular the entire condition (59) is preserved. It is essential to include tQ: for $a > 1$, cancellation of $C'(1)$ alone would leave $Q \log a$.

Subtract the corrected targets. Since $(T_m S)(1/2) = 0$, the inverse first moment vanishes, and the $1/(2n)$ terms cancel in their difference. Also $(T_m S)(1) = (T_m S)'(1) = 0$, so all gamma and logarithmic terms cancel when (15) is summed over its coefficients. For $z > 0$, the exact finite expansion is

$$\Phi(z) = \sum_{k=0}^{m-1} (-1)^k \mu_k z^{-k-1} + (-1)^m z^{-m} \int \frac{u^m}{u+z}\, d\mu(u).$$

For $m = 0$ the sum is empty. The remainder has absolute value at most $\mu_m z^{-m-1}$. Each of the first m terms vanishes in the coefficient sum because $(T_m S)(4^{-k}) = 0$ for $1 \le k \le m$. Thus

$$|\Delta| \le \frac{\mu_m a^{-2m-2}}{|Q|} (T_m S)^{\mathrm{abs}}(x_0) \le \frac{\mu_m a^{-2m-2}}{|Q|} T_m^{\mathrm{abs}}(x_0) S^{\mathrm{abs}}(x_0),$$

which is (61).

All surviving indices satisfy $a2^i \le N_0$. Each corrected harmonic denominator divides D_0, and 2^E clears all r_i. Hence Z is an integer, and reduction of $Z/(Q2^E D_0)$ gives (62). The odd-part assertion follows immediately. Finally $\gamma - r_R = (\gamma - r_C) + \Delta$, so (63) follows from the triangle inequality and the exact value of q_R. □

An infinite shared-prime cancellation with positive errors. Set the external base $a = 1$, take $m = 0$, and put

$$C_t(x) = 1 + x^t T_0(x), \qquad t \ge 1.$$

The remainder is $R = 1$. Its corrected target is exactly $H_1 - 1/2 = 1/2$, with final denominator 2. The original target is

$$r_{C_t} = \frac{1}{2} + V_t, \qquad V_t = \sum_{i=0}^{3} [x^i] T_0(x) H_{2^{t+i}}.$$

Since $T_0^{\mathrm{abs}}(1/4) = 75/32$, the finite Binet remainder gives

$$|V_t| \le \frac{25}{128\,4^t} < \frac{3}{40}.$$

On the other hand $h_1(u) > 1 - u$ for $u > 0$, so

$$\gamma - \frac{1}{2} > \mu_0 - \mu_1 = \frac{3}{40}.$$

Every original target and every compressed target therefore has a strictly positive gamma error.

For any prime ℓ in $(2^{t+2}, 2^{t+3}]$, only the last harmonic sum in V_t contains a denominator divisible by ℓ, and its coefficient is 2. It contains exactly the single term $1/\ell$ with such a denominator. Thus $v_\ell(r_{C_t}) = -1$. All these primes disappear after compression. For instance the exact reduced denominators for $t = 1, 2$ are respectively 90090 and 9025222055850 before compression, and 2 afterward.

This example explicitly removes a block of primes already present in the base. It lies outside an obstruction whose hypothesis requires that the base avoid that block. At the same time, it changes the target to $1/2$, leaving the error $\gamma - 1/2$ fixed. It therefore proves the arithmetic cancellation mechanism without claiming the missing small-error estimate needed for irrationality.

12 Rational localization at the double zero of the positive weight

The positive square weight $(t-1)\log t$ vanishes quadratically at $t = 1$. This section constructs rational polynomials localized at points tending to that zero. The coefficients depend on Laguerre values and lie outside the fixed-polynomial-schedule obstruction. The real integral becomes small with an explicit order, and its full primitive cost is expressed in terms of the actual reduced denominator. A one-step change of the cutoff also removes a specific adverse prime factor exactly.

Let $L_k^{(1)}$ denote the generalized Laguerre polynomial with parameter 1, and let $L_k = L_k^{(0)}$. For an integer $M \ge 2$ and $\sigma \in \{0, -1\}$, define

$$N = M^2 + \sigma, \qquad x = 1 - \frac{1}{M}, \qquad H = N + 1, \qquad w_k = \left(1 - \frac{k}{H}\right)^2 \quad (0 \le k \le N),$$

$$p_{N,x}(t) = \sum_{k=0}^{N} \frac{w_k L_k^{(1)}(x) L_k^{(1)}(t)}{k+1}, \qquad Q_M(t) = t\, p_{N,x}(t). \tag{64}$$

These polynomials have rational coefficients, constructed without using a value or an assumption about the arithmetic nature of γ. Write

$$I_M = A_M - \gamma B_M = \int_0^\infty Q_M(t)^2 (t-1)\log t\, e^{-t} dt > 0.$$

The positivity is strict because Q_M is nonzero. Indeed its ordinary Laguerre coefficient at index zero is $w_0 L_0^{(1)}(x) = 1$.

Theorem 12.1 (Exact rational certificates for the localized family)**.** *Use the inner product* $\langle f, g\rangle_1 = \int_0^\infty f(t)g(t)te^{-t}dt$ *and put*

$$H_j = \langle p_{N,x}, t^j p_{N,x}\rangle_1, \qquad T_M = H_2 + H_1, \qquad U_M = H_3 - H_2 - H_1 + H_0.$$

Then all these numbers are rational, $B_M = H_2 - H_1$*, and*

$$\frac{2B_M^2}{T_M} \le I_M \le \frac{U_M}{2}. \tag{65}$$

When $B_M \neq 0$, let q_M be the actual reduced denominator of A_M/B_M. The complete primitive error $\mathcal{E}_M$ satisfies the entirely rational certificate

$$\frac{2q_M|B_M|}{T_M} \leq \mathcal{E}_M \leq \frac{q_M U_M}{2|B_M|}. \tag{66}$$

For exact integerization, put

$$F_M = H^2 N! M^N, \qquad D_N = \frac{\operatorname{lcm}(1,\dots,N+1)}{2}, \qquad S_M = D_N F_M^2.$$

Then $S_M A_M, S_M B_M$ are integers. If $g_M = \gcd(S_M A_M, S_M B_M) > 0$, the identities

$$\mathcal{E}_M = \frac{S_M I_M}{g_M}, \qquad q_M = \frac{S_M|B_M|}{g_M} \quad (B_M \neq 0)$$

use the complete gcd, not merely a chosen clearing factor.

Proof. For every $t > 0$,

$$\frac{2(t-1)^2}{t+1} \leq (t-1)\log t \leq \frac{(t-1)^2(t+1)}{2t}.$$

For the lower inequality use $\log t = 2\operatorname{artanh}((t-1)/(t+1))$. For the upper one, the derivative of $(t - t^{-1})/2 - \log t$ is $(t-1)^2/(2t^2) \geq 0$, and its value at 1 is zero. Multiplication by $Q_M(t)^2 = t^2 p_{N,x}(t)^2$ and integration gives the upper bound $U_M/2$. Cauchy–Schwarz gives

$$B_M^2 \leq \left(\int_0^\infty \frac{t(t-1)^2}{t+1} p_{N,x}(t)^2 t e^{-t} dt\right)\left(\int_0^\infty t(t+1) p_{N,x}(t)^2 t e^{-t} dt\right) \leq \frac{I_M T_M}{2}.$$

This proves (65). Since $I_M > 0$,

$$\mathcal{E}_M = q_M \left|\gamma - \frac{A_M}{B_M}\right| = q_M \frac{I_M}{|B_M|},$$

which gives (66).

The elementary identity

$$t L_k^{(1)}(t) = (k+1)(L_k(t) - L_{k+1}(t))$$

shows that, with $z_k = w_k L_k^{(1)}(x)$,

$$Q_M = \sum_{k=0}^{N} z_k (L_k - L_{k+1}).$$

Its ordinary Laguerre coefficient vector is $(z_0, z_1 - z_0, \dots, z_N - z_{N-1}, -z_N)$. The finite coefficient formula for $L_k^{(1)}(x)$ shows that F_M clears this vector. Lemma 10.1, applied to an integer Laguerre vector of length $N+2$, proves that S_M clears (A_M, B_M). The remaining identities follow by complete reduction of this pair. □

Theorem 12.2 (Uniform localization and its exact arithmetic threshold)**.** *For either $\sigma = 0$ or $\sigma = -1$ in (64), let*

$$\kappa = \frac{128e}{315\pi} > 0.$$

Then, as $M \to \infty$,

$$B_M = -\kappa + O\left(\frac{\log M}{M}\right), \qquad T_M = 2\kappa M + O(\log M), \qquad U_M = O(M^{-1}).$$

Consequently

$$I_M = \Theta(M^{-1}), \qquad \gamma - \frac{A_M}{B_M} = \Theta(M^{-1})$$

with the second quantity positive for all sufficiently large M. For the actual reduced denominator,

$$\mathcal{E}_M = \Theta\left(\frac{q_M}{M}\right). \tag{67}$$

Thus this particular family would satisfy primitive-error decay exactly if $q_M = o(M)$; equivalently, $Mg_M/S_M \to \infty$. Neither denominator condition is assumed or proved here.

Proof. All estimates below are uniform for x in the compact interval $[1/2, 1]$. We use the established Laguerre asymptotic [8]

$$L_k^{(\alpha)}(x) = \frac{e^{x/2}k^{\alpha/2-1/4}}{\sqrt{\pi}\,x^{\alpha/2+1/4}}\left[\cos\left(2\sqrt{kx} - \left(\frac{\alpha}{2}+\frac{1}{4}\right)\pi\right) + O(k^{-1/2})\right] \tag{68}$$

for $\alpha = 0, 1$. Its compact-x uniformity is essential because $x = 1 - 1/M$ varies with M.

Let $\phi_k = L_k^{(1)}/\sqrt{k+1}$, an orthonormal basis for the inner product $\langle\ ,\ \rangle_1$. Multiplication by t has Jacobi matrix

$$J_{kk} = 2k + 2, \qquad J_{k,k+1} = -\sqrt{(k+1)(k+2)}.$$

The coefficient vector of $p_{N,x}$ is $c_k = w_k L_k^{(1)}(x)/\sqrt{k+1}$ for $k \le N$, and zero otherwise. Write

$$S = \|c\|^2, \qquad g = (J - x)c, \qquad \delta = \langle c, g\rangle.$$

First, squaring (68) for $\alpha = 1$ gives

$$S = \kappa(x)\sqrt{N} + O(\log(N+2)), \qquad \kappa(x) = \frac{128e^x}{315\pi x^{3/2}}. \tag{69}$$

Here the nonoscillating sum is

$$\frac{e^x}{2\pi x^{3/2}}\sum_{k=1}^{N} k^{-1/2}\left(1 - \frac{k}{N+1}\right)^4,$$

whose leading coefficient is

$$\frac{e^x}{2\pi x^{3/2}}\int_0^1 u^{-1/2}(1-u)^4 du = \frac{128e^x}{315\pi x^{3/2}}.$$

The asymptotic remainders sum to $O(\log(N+2))$. For the oscillating sum, comparison with its integral has the same error bound: the phase derivative gives an integrable-on-finite-interval bound $O(1/t)$ for the derivative of the summand. Substitution $t = y^2$ bounds the integral by $O(1)$ using integration by parts. The frequency $4\sqrt{x}$ stays bounded away from zero, so all constants are uniform.

Second, the exact Jacobi localization identity is

$$\delta = \sum_{k=0}^{N} L_k^{(1)}(x)L_{k+1}^{(1)}(x)(w_k - w_{k+1})^2, \qquad w_{N+1} = 0.$$

The point bounds from (68) and $|w_k - w_{k+1}| \le 2/(N+1)$ imply

$$|\delta| = O(N^{-1/2}). \tag{70}$$

No sign of the individual products is used.

We also need a weighted residual bound, not just weak concentration. Put $H = N + 1$, $L_{-1}^{(1)} = 0$, and

$$U_k = L_k(x) + L_{k+1}(x), \qquad V_k = L_{k-1}^{(1)}(x) + L_{k+1}^{(1)}(x).$$

Direct substitution of the quadratic taper gives, for $0 \le k \le N$,

$$g_k = \sqrt{k+1}\left[\frac{2(H-k)}{H^2}U_k - \frac{1}{H^2}V_k\right], \qquad g_{N+1} = -\frac{\sqrt{N+2}}{H^2}L_N^{(1)}(x).$$

All later coefficients vanish. The identities

$$L_{k+1}^{(1)} - L_k^{(1)} = L_{k+1}, \qquad L_{k+1} - L_k = -\frac{x}{k+1}L_k^{(1)}$$

and (68) give

$$U_k = O((k+1)^{-1/4}), \quad U_{k+1} - U_k = O((k+1)^{-3/4}),$$
$$V_k = O((k+1)^{1/4}), \quad V_{k+1} - V_k = O((k+1)^{-1/4}).$$

It follows that $g_k = O((k+1)^{1/4}/H)$ in the bulk, while $g_N, g_{N+1} = O(H^{-5/4})$ at the endpoint. Thus $\|g\|^2 = O(N^{-1/2})$.

For the additional factor of J, use the exact nonnegative energy identity

$$\langle g, Jg\rangle = g_0^2 + \sum_{k\ge 0}\left(\sqrt{k+1}\,g_k - \sqrt{k+2}\,g_{k+1}\right)^2.$$

The displayed difference bounds show that the bulk differences are $O((k+1)^{1/4}/H)$; the two endpoint contributions are $O(H^{-3/4})$ before squaring. Therefore

$$\langle g, (J+1)g\rangle = O(N^{-1/2}). \tag{71}$$

This controls the required unbounded observable on the full half-line.

The exact moment identities now give

$$B_M = (x^2 - x)S + (2x-1)\delta + \|g\|^2,$$

$$T_M = (x^2 + x)S + (2x+1)\delta + \|g\|^2.$$

Since $\sqrt{N} = M + O(M^{-1})$, $x = 1 - M^{-1}$, and $\kappa(1) = \kappa$, these prove the stated asymptotics of B_M and T_M. Finally,

$$U_M = \langle (J-1)c, (J+1)(J-1)c\rangle \le 2\langle g, (J+1)g\rangle + 2(x-1)^2\langle c, (J+1)c\rangle = O(M^{-1}).$$

The rational sandwich proves $I_M = \Theta(M^{-1})$: its lower bound is asymptotic to κ/M, and its upper bound is $O(M^{-1})$. Since $|B_M| \to \kappa$, both remaining assertions follow. □

Theorem 12.3 (An adverse prime factor and its removal by one cutoff step)**.** *For the family above:*

1. *If $\sigma = 0$ and $M = p^e$ is any prime power, then*

$$v_p(A_M/B_M) = -2e.$$

Thus the actual reduced denominator contains M^2. Along every prime-power sequence $M \to \infty$, the primitive error is bounded below by a positive constant times M.

2. *If $\sigma = -1$ and $M = p^e$ with p odd, then*

$$v_p(q_M) \le 2e - 2.$$

For $p \ge 5$ this improves to

$$v_p(q_M) \le \max(0, 2e - 3).$$

In particular, at $M = p$ every odd prime p, including $p = 3$, is absent from the actual reduced denominator.

These statements concern the indicated prime factor only. The second case does not bound the cost of all remaining denominator factors.

Proof. Put $u_k = L_k - L_{k+1}$ and $z_k = w_k L_k^{(1)}(x)$, so $Q_M = \sum_{k=0}^{N} z_k u_k$. For $M = p^e$, $v_p(x) = -e$. The finite Laguerre formula gives

$$L_k^{(1)}(x) = \frac{(-x)^k}{k!}(1 + p^e \mathbb{Z}_p).$$

Every nonleading term divided by the leading term is an integer times x^{-j} for some $j \ge 1$. In particular the displayed leading unit is nonzero. Consequently, for $1 \le j \le N$,

$$v_p\left(\frac{z_{N-j}}{z_N}\right) = je + v_p\left(\frac{N!}{(N-j)!}\right) + 2v_p(j+1). \tag{72}$$

The ordinary Laguerre Gram matrices give exactly

$$B(u_N) = 6N + 4, \tag{73}$$

$$A(u_N) = (6N + 4)H_N + 7N + 8, \tag{74}$$

$$2\mathcal{A}(u_N, u_{N-1}) = -2(4N + 1)H_N - \frac{32N + 7}{3}. \tag{75}$$

Here $\mathcal{A}$ denotes the bilinear form associated with A.

First take $N = p^{2e}$. Every ratio in (72) has valuation at least $3e$. Since $H_{p^{2e}}$ has the unique summand $1/p^{2e}$ of lowest p-adic valuation,

$$v_p(H_{p^{2e}}) = -2e.$$

For odd p, (73) is a unit and (74) has valuation $-2e$. For $p = 2$, their valuations are respectively 2 and $2 - 2e$.

Write $Q_M/z_N = u_N + \sum_{j\ge1}(z_{N-j}/z_N)u_{N-j}$. In integer Laguerre coordinates of degree $N + 1$, the denominators of quadratic coefficients divide $\operatorname{lcm}(1, \dots, N + 1)/2$. Its valuation is $2e$ for odd p and $2e - 1$ for $p = 2$. Thus the perturbation in A has valuation at least e for odd p and $e + 1$ for $p = 2$. The perturbation in B has valuation at least $3e$. Both are strictly higher than the corresponding leading valuations. Division gives $v_p(A_M/B_M) = -2e$ in every case. The real lower bound then follows from (67).

Now take $N = p^{2e} - 1$ with p odd. Pairing j and $p^a - j$ in the sum over p-units shows

$$H_{p^a-1} = \frac{1}{p} H_{p^{a-1}-1} + V_a, \qquad v_p(V_a) \ge a.$$

Also $v_p(H_{p-1}) \ge 1$. Hence

$$v_p(H_N) \ge 2 - 2e.$$

For $p \ge 5$, pairing modulo p^2 and using $\sum_{j=1}^{p-1} j^{-2} \equiv 0 \pmod p$ gives $v_p(H_{p-1}) \ge 2$, improving this to $v_p(H_N) \ge 3 - 2e$.

The leading value $B(u_N) = 6p^{2e} - 2$ is a p-adic unit. Every ratio in (72) has valuation at least je, so all changes to B are divisible by p. Thus the normalized B remains a unit.

For A, all linear perturbation terms with $j \geq 2$ and all quadratic perturbation terms are p-integral, because their numerator valuations are at least $2e$ and the global quadratic coefficient denominators have valuation at most $2e$. Only $j = 1$ needs a local check. For $p \geq 5$, (75) has valuation at least $\min(0, 3 - 2e)$, and multiplication by z_{N-1}/z_N, of valuation e, cannot worsen that bound.

For $p = 3$, $v_3(4N + 1) = 1$. The adjacent coefficient in (75) therefore has valuation at least $\min(3-2e, -1)$. After multiplication by the ratio, its valuation is at least $\min(3-e, e-1) \geq 2-2e$. This includes $p = 3, e = 1$. The leading value (74) satisfies the same required lower bounds. Since the normalized B is a unit, the stated bounds for the reduced denominator follow. □

What the one-step repair accomplishes. The two cutoffs have the same rigorously proved real error order $\Theta(1/M)$. At $M = p$, changing $N = p^2$ to $N = p^2 - 1$ removes the entire selected prime-power denominator p^2 rather than merely changing a clearing factor. The exact checks give the following comparison:

$M = p$	digits of q $(N = p^2)$	digits of q $(N = p^2 - 1)$	$v_p(q)$ $(N = p^2)$	$v_p(q)$ $(N = p^2 - 1)$
3	20	17	2	0
5	86	79	2	0
7	184	177	2	0

The remaining denominators in these examples are large. No estimate $q_M = o(M)$ is inferred from this local improvement.

12.1 The repaired cutoff still fails on two fixed-prime towers

The cutoff change $M^2 \mapsto M^2 - 1$ removes the selected prime entirely when M itself is an odd prime. It does not have that effect along prime-power towers with a fixed prime. The following additional consequence gives exact valuations on two such towers. It is a further obstruction within the present construction, not a claim that every modified localization schedule fails.

Proposition 12.4 (Exact denominator costs after repair on the 3- and 5-power towers)**.** *Use the localized family with $\sigma = -1$, so $N = M^2 - 1$, and let q_M be the actual reduced denominator of A_M/B_M. Then, for every integer $e \geq 1$,*

$$M = 3^e : \qquad v_3(q_M) = 2e - 2,$$
$$M = 5^e : \qquad v_5(q_M) = \max(0, 2e - 3).$$

In particular,

$$\frac{q_M}{M} \geq \frac{M}{9} \quad (M = 3^e), \qquad \frac{q_M}{M} \geq \frac{M}{125} \quad (M = 5^e, e \geq 2).$$

Together with Theorem 12.2, these bounds imply that the primitive errors tend to $+\infty$ on both fixed-prime towers, even with the repaired cutoff.

Proof. First we determine the relevant harmonic valuations exactly. For an odd prime p and an integer $a \geq 2$, write

$$H_{p^a-1} = \frac{1}{p} H_{p^{a-1}-1} + V_{p,a}, \qquad V_{p,a} = \sum_{\substack{1 \leq j < p^a \\ p \nmid j}} \frac{1}{j}.$$

Pairing j with $p^a - j$ gives

$$\frac{1}{j} + \frac{1}{p^a - j} = \frac{p^a}{j(p^a - j)},$$

so $v_p(V_{p,a}) \geq a$. The exact initial values $H_2 = 3/2$ and $H_4 = 25/12$ now imply, by induction,

$$v_3(H_{3^a-1}) = 2 - a, \qquad v_5(H_{5^a-1}) = 3 - a \qquad (a \geq 1). \tag{76}$$

Indeed, at each step $a \geq 2$, the inherited valuations are $2 - a$ and $3 - a$, respectively, both strictly smaller than the unit-block lower bound a. Thus no cancellation can change the leading valuation.

We next check that these harmonic valuations survive the complete quadratic form. Use the notation

$$u_k = L_k - L_{k+1}, \qquad z_k = \left(\frac{N+1-k}{N+1}\right)^2 L_k^{(1)}\left(1 - \frac{1}{M}\right).$$

For $M = p^e$, the leading-term valuation of the Laguerre polynomial shows $z_N \neq 0$. Normalize the polynomial by this coefficient:

$$\frac{Q_M}{z_N} = u_N + \sum_{j=1}^{N} t_j u_{N-j}, \qquad t_j = \frac{z_{N-j}}{z_N}.$$

The exact coefficient estimate proved in Theorem 12.3 is

$$v_p(t_j) = je + v_p\left(\frac{N!}{(N-j)!}\right) + 2v_p(j+1) \geq je. \tag{77}$$

For $j = 1$ and either $p = 3$ or $p = 5$, equality is $v_p(t_1) = e$, because $N = p^{2e} - 1$ and 2 are p-adic units.

The exact leading coefficients are

$$B(u_N) = 6N + 4 = 6p^{2e} - 2, \qquad A(u_N) = (6p^{2e} - 2)H_N + 7p^{2e} + 1.$$

The first is a p-adic unit for both primes. For $e \geq 2$, (76) gives the strictly negative valuations

$$v_3(A(u_N)) = 2 - 2e, \qquad v_5(A(u_N)) = 3 - 2e. \tag{78}$$

There is no cancellation within this leading expression: the harmonic term has negative valuation, while $7p^{2e} + 1$ is a unit.

All coefficients of the quadratic form B in integer Laguerre coordinates are integers. Therefore every change from $B(u_N)$ after substituting (77) is divisible by p, and

$$v_p(B(Q_M/z_N)) = 0.$$

For A, the half-lcm clearing factor in degree $N + 1 = p^{2e}$ shows that every quadratic coefficient has valuation at least $-2e$ for these odd primes. Consequently, every linear perturbation $t_j\, 2\mathcal{A}(u_N, u_{N-j})$ with $j \geq 2$ has valuation at least $je - 2e \geq 0$, and every quadratic perturbation in the t_j has valuation at least $(j + k)e - 2e \geq 0$.

It remains to check the single adjacent linear perturbation. Its coefficient is exactly

$$C_N := 2\mathcal{A}(u_N, u_{N-1}) = -2(4N + 1)H_N - \frac{32N + 7}{3}.$$

For $p = 3$, the factor $4N + 1 = 4 \cdot 3^{2e} - 3$ has valuation one, while $32N + 7 = 32 \cdot 3^{2e} - 25$ is a unit. Hence

$$v_3(t_1 C_N) \geq e + \min(3 - 2e, -1) = \min(3 - e, e - 1) > 2 - 2e \qquad (e \geq 2).$$

For $p = 5$, $4N + 1$ and 3 are units, and (76) gives

$$v_5(t_1 C_N) \geq e + (3 - 2e) = 3 - e > 3 - 2e \qquad (e \geq 2).$$

Thus every perturbation has valuation strictly greater than the corresponding negative leading valuation in (78). The full normalized A retains that valuation. Since the normalized B is a unit and quadratic homogeneity cancels z_N^2 in A_M/B_M, this proves

$$v_3(A_M/B_M) = 2 - 2e, \qquad v_5(A_M/B_M) = 3 - 2e \qquad (e \geq 2).$$

Reduction of a rational number makes its denominator valuation the negative part of its valuation, giving the asserted exponents for $e \geq 2$. At $e = 1$, the odd-prime removal conclusion of Theorem 12.3 gives exponent zero in both cases, which is exactly the formula stated here.

The resulting divisibilities are $M^2/9 \mid q_M$ when $M = 3^e$, and $M^2/125 \mid q_M$ when $M = 5^e$ and $e \geq 2$. They imply the displayed bounds for q_M/M. Finally the established estimate $\mathcal{E}_M = \Theta(q_M/M)$ gives divergence of the primitive error on both towers. □

Small exact checks and their role. The leading-unit checks at $e = 2$ reduce to the finite rational sums H_{80} and H_{624}. Exact arithmetic gives

(p, M, N)	$v_p(H_N)$	$p^{-v_p(A(u_N))}A(u_N) \pmod p$	$B(u_N) \pmod p$
$(3, 9, 80)$	-2	2	1
$(5, 25, 624)$	-1	4	3

The perturbation bounds above show that the same scaled A residues hold for $A(Q_M/z_N)$, and that the B residues also remain unchanged. Thus these checks certify denominator exponents 2 and 1, respectively. The all-index assertions follow from the valuation proof, not extrapolation from these checks.

12.2 A fresh-prime test after the cutoff repair

The removal of a prime dividing the localization parameter does not control primes of the size of the cutoff. The following exact test detects many such primes in the *actual* reduced denominator. Here $A(Q)$ and $B(Q)$ have their positive-square meaning, $A(Q) - \gamma B(Q) = \int_0^\infty Q(t)^2 (t-1) \log t \, e^{-t} \, dt$.

Proposition 12.5 (Fresh-prime denominator test)**.** *Let $M \geq 2$, $H = M^2$, $N = H - 1$, $x = (M-1)/M$, and*

$$w_k = \left(\frac{H-k}{H}\right)^2, \qquad Q(t) = \sum_{k=0}^{N} w_k L_k^{(1)}(x)\big(L_k(t) - L_{k+1}(t)\big).$$

Let ℓ be an odd prime with $H/2 < \ell \leq H$, and set $s = N - \ell$. Define the truncated polynomial

$$W_{M,\ell}(t) = \sum_{r=0}^{s} w_r L_r^{(1)}(x)\big(L_r(t) - L_{r+1}(t)\big), \qquad R_{M,\ell} = x^2 B(W_{M,\ell}) \pmod \ell.$$

Then $s \geq 0$, $s \leq \ell - 2$, the residue on the right is defined, and the two numbers on the left of

$$\ell^3 A(Q) \equiv \ell^2 B(Q) \equiv R_{M,\ell} \pmod \ell \tag{79}$$

are ℓ-integral. If $R_{M,\ell} \neq 0$, then

$$v_\ell(A(Q)) = -3, \qquad v_\ell(B(Q)) = -2, \qquad v_\ell\left(\operatorname{den} \frac{A(Q)}{B(Q)}\right) = 1.$$

In particular, if $B(Q) \neq 0$, the completely computable integer

$$G_M = \prod_{\substack{\ell \text{ odd prime},\ H/2<\ell\le H \\ R_{M,\ell}\neq 0}} \ell$$

divides the actual reduced denominator q_M.

Proof. Since $H = M^2$ is composite, $\ell \le H-1$, and $H < 2\ell$ gives $0 \le s \le \ell-2$. Such a prime divides neither M nor $M-1$, so x, H are ℓ-adic units. The finite Laguerre formula, Wilson's theorem, Lucas's congruence, and $x^\ell \equiv x$ give

$$\ell L_{\ell+r}^{(1)}(x) \equiv x L_r^{(1)}(x) \pmod{\ell} \qquad (0 \le r \le s). \tag{80}$$

For completeness, in $L_n^{(1)}(x) = \sum_{j=0}^{n}(-1)^j\binom{n+1}{j+1}x^j/j!$, the terms $j < \ell$ disappear after multiplication by ℓ. Writing $j = \ell + a$ in the remaining terms gives

$$\frac{\ell}{(\ell+a)!} \equiv -\frac{1}{a!}, \qquad \binom{\ell+r+1}{\ell+a+1} \equiv \binom{r+1}{a+1},$$

which proves (80).

Write $z_k = w_k L_k^{(1)}(x)$ and let c be the ordinary Laguerre coefficient vector of Q:

$$c = (z_0, z_1 - z_0, \dots, z_N - z_{N-1}, -z_N).$$

The vector $\widetilde{c} = \ell c$ is ℓ-integral, vanishes modulo ℓ in indices $0, \dots, \ell-1$, and in indices $\ell, \dots, H$ is congruent to x times the *complete* ordinary Laguerre coefficient vector of $W_{M,\ell}$, including its final coefficient $-z_s$. This uses $w_{\ell+r} \equiv w_r$. The exact Gram form

$$B(c) = \sum_{i=0}^{H} 2ic_i^2 - 2\sum_{i=0}^{H-1}(i+1)c_ic_{i+1}$$

therefore gives $B(\widetilde{c}) \equiv x^2 B(W_{M,\ell})$.

All entries of ℓA are ℓ-integral because $H < 2\ell$. For indices $i, j \ge \ell$, its diagonal residues are $2i$, its adjacent cross-term residues are $-2(i+1)$, and its remaining cross-term residues vanish. Indeed $\ell H_i \equiv 1$ for $\ell \le i \le H$ and

$$2A_{ij} = \frac{2(i+1)}{j-i-1} - \frac{4i}{j-i} + \frac{2i}{j-i+1} \qquad (j-i \ge 2).$$

The only possible exceptional gap within the high block is $j - i = \ell - 1$. It requires $H = 2\ell - 1$, $i = \ell$, and $j = 2\ell - 1$; the potential residue contains $i \equiv 0$ and is zero. All terms involving a low coefficient vanish because that coefficient is divisible by ℓ. Consequently $\ell A(\widetilde{c}) \equiv B(\widetilde{c})$. Quadratic homogeneity proves (79), and a nonzero residue gives the three exact valuations. □

Scope and a zero-residue example. The nonzero-residue condition is sufficient, and is not necessary. For $M = 5$, $N = 24$, $\ell = 17$, $s = 7$, one has $x \equiv 11 \pmod{17}$ and $B(W_{5,17}) \equiv 0 \pmod{17}$. Exact rational arithmetic nevertheless gives

$$v_{17}(A(Q)) = -2, \qquad v_{17}(B(Q)) = 0, \qquad v_{17}(q_5) = 2.$$

Thus a zero residue does not certify cancellation of the prime. The accompanying verifier checks the full rational forms and actual reduced denominators for $2 \le M \le 11$. Its separate finite-field scan over $2 \le M \le 30$ has 786 eligible pairs, of which 781 have nonzero residue. The five zero pairs (M, ℓ) are $(5, 17)$, $(17, 197)$, $(19, 241)$, $(23, 487)$, and $(24, 449)$. These finite data do not establish a lower bound for G_M as $M \to \infty$. Combined with the proved localization estimate, the proposition gives $\mathcal{E}_M \gg G_M/M$ for sufficiently large M; it supplies no unconditional global growth rate for G_M, and no irrationality proof.

13 Exact logarithm cancellation and the nonvanishing barrier

This section changes the construction. It obtains small log-free integer forms from a dense block of proper kernels. Positivity is not preserved by the counting argument. We prove that the same argument also produces formal-zero combinations, and then give a fixed-degree positive construction that would work if a specific independent-pair height bound were established. That height bound is not proved here.

For $n \geq 1$, define

$$R_n(u) = \frac{\prod_{k=0}^{2n-1}(4u+k^2)}{\prod_{j=n}^{3n}(u+j^2)}, \qquad I_n = \int_0^\infty R_n(u)\, d\mu(u) > 0.$$

The residues, constant coefficient, and prime-log coefficients are

$$A_{n,j} = (-1)^{j-n}\frac{4j^2(2j+2n-1)!(n+j-1)!}{(2j-2n)!(j-n)!(3n-j)!(3n+j)!},$$
$$p_n = \sum_{j=n}^{3n} A_{n,j}\left(H_j - \frac{1}{2j}\right), \qquad E_{n,\ell} = \sum_{j=n}^{3n} A_{n,j} v_\ell(j).$$

There is no polynomial part in the partial fraction decomposition, and $\sum_j A_{n,j} = 16^n$. Thus Binet's formula gives exactly

$$I_n = 16^n\gamma - p_n + \sum_{\ell \leq 3n} E_{n,\ell}\log \ell. \tag{81}$$

Only primes occur in the last sum.

Lemma 13.1 (Uniform height and raw integral bounds)**.** *For every $n \geq 1$ and $n \leq j \leq 3n$,*

$$2A_{n,j} \in \mathbb{Z}, \qquad 2|A_{n,j}| \leq 6\,2^{12n}.$$

Moreover

$$0 < I_n < U_n := \frac{(4n)!((n-1)!)^2}{8\,16^n((3n)!)^2}, \qquad U_{n+1} < U_n, \tag{82}$$

and, with $\kappa_0 = \log(729/16)$,

$$\log U_n = -\kappa_0 n + O(\log(n+1)).$$

Proof. Put $a = j+n$ and $b = j-n$. With

$$B = \binom{2a}{b}, \quad C = \binom{a}{2b}, \quad B' = \binom{2a-1}{b-1}, \quad C' = \binom{a-1}{2b-1},$$

where $B' = C' = 0$ at $b = 0$, direct factorial simplification gives

$$2|A_{n,j}| = BC + 4B'C + B'C'.$$

This proves integrality. Every product in this expression is at most $2^{3a} \leq 2^{12n}$, proving the stated height bound.

For the integral estimate, put $d\nu(u) = du/(2\sinh(2\pi\sqrt{u}))$. Then $1 < d\mu/d\nu = 1+e^{-2\pi\sqrt{u}} < 2$. Let $T_j(u) = P_{2j}^{(1)}(2\sqrt{u};\pi/2)$ be the even Meixner–Pollaczek polynomials in their usual normalization. Their generating function and orthogonality [9, 10] imply

$$S_r(u) := \sum_{j=0}^{r}\binom{r}{j}T_j(u) = \frac{4^r}{(2r)!}\prod_{k=0}^{r-1}(4u+k^2), \qquad \int_0^\infty S_r\, d\nu = \frac{1}{16}.$$

For clarity, the generating function of the binomial transform is $\cosh(4\sqrt{u}\arcsin\sqrt{z})$, whose coefficient gives the displayed product. In the integral, all T_j with $j>0$ vanish by orthogonality; the constant norm is $\int d\nu = 1/16$. The product is positive for $u>0$, so $\int S_r d\mu < 1/8$. Since

$$R_n(u) = \frac{(4n)!}{16^n}\frac{S_{2n}(u)}{\prod_{j=n}^{3n}(u+j^2)},$$

replacing each denominator factor by j^2 gives (82). A direct quotient estimate gives

$$\frac{U_{n+1}}{U_n} = \frac{(4n+1)(4n+2)(4n+3)(4n+4)n^2}{16(3n+1)^2(3n+2)^2(3n+3)^2} \le \frac{16}{729}\left(1+\frac{1}{n}\right)^4 \le \frac{256}{729} < 1.$$

Finally Stirling's formula gives the logarithmic estimate. □

Theorem 13.2 (Thin-block cancellation with an exponential bound, possibly zero)**.** *For integers $N \ge 2$, define using integer operations*

$$h_N = \max\left(1, \left\lfloor \sqrt{\lfloor \log_2 N \rfloor} \right\rfloor\right), \quad M = N + \left\lceil \frac{N}{h_N} \right\rceil, \quad d = M - N + 1,$$

$$D_N = 4\,\mathrm{lcm}(1,\dots,3M), \qquad \delta_* = \log(729/16) - 3 > 0.$$

For every sufficiently large N, there is a nonzero integer vector $c = (c_N,\dots,c_M)$ satisfying

$$\sum_{n=N}^{M} c_n E_{n,\ell} = 0 \quad \text{for every prime } \ell, \qquad \log\max(1,\|c\|_\infty) = O\!\left(\frac{N}{\sqrt{\log N}}\right).$$

Its exactly log-free form is

$$F_N = D_N \sum_{n=N}^{M} c_n I_n = Q_N\gamma - P_N, \qquad Q_N, P_N \in \mathbb{Z}, \tag{83}$$

and satisfies

$$|F_N| \le \exp(-\delta_* N + o(N)). \tag{84}$$

This is an upper bound; F_N is allowed to be exactly zero.

Indeed, for every sufficiently large N, there also exists a nonzero integer vector of the same height order for which all prime-log coefficients, Q_N, and P_N are zero simultaneously. Its rational kernel $\sum c_n R_n$ is nonzero and changes sign on $(0,\infty)$.

Proof. Use the integer log matrix

$$V_{\ell,n} = 2E_{n,\ell}, \qquad \ell \le 3M \text{ prime}, \quad N \le n \le M.$$

It has $r = \pi(3M)$ rows and d columns, and its entries have absolute value at most the integer

$$H_N = 6(2M+1)\lceil \log_2(3M)\rceil 2^{12M}.$$

The prime number theorem [7] and the chosen block give

$$M/N \to 1, \quad d \asymp \frac{N}{\sqrt{\log N}}, \quad r \asymp \frac{N}{\log N}, \quad \log H_N = O(N).$$

Thus $d > r$ for all sufficiently large N.

Here is the full counting argument. For any r-by-d integer matrix with entries bounded by H, let K be a positive integer such that

$$K^{d-r} > (3dH)^r.$$

The $(K+1)^d$ vectors in $\{0,\dots,K\}^d$ have images with at most $(2dKH+1)^r \le (3dKH)^r < K^d$ possible values. Two images coincide. Subtracting the input vectors gives a nonzero integer null vector of supremum norm at most K. Taking the smallest admissible integer K yields

$$\log K \le \frac{r}{d-r}\log(3dH) + O(1) = O\Big(\frac{N}{\sqrt{\log N}}\Big).$$

This proves exact log cancellation and the height estimate. It also gives a finite, though not efficient, construction by enumeration.

Since $2A_{n,j}$ is integral, every denominator in p_n divides $4\,\mathrm{lcm}(1,\dots,3M) = D_N$. Thus the coefficients

$$Q_N = D_N \sum c_n 16^n, \qquad P_N = D_N \sum c_n p_n$$

in (83) are integers. Positivity and monotonicity of the raw bounds give

$$|F_N| \le D_N dKU_N.$$

The same prime number theorem gives $\log D_N = 3M + o(M) = 3N + o(N)$, while $\log(dK) = o(N)$. Lemma 13.1 therefore proves (84). The margin is positive since $e < 3$ gives $e^3 < 27 < 729/16$.

To see exactly why nonvanishing is missing, append to V the two integer rows $(16^n)_{n=N}^{M}$ and $(D_N p_n)_{n=N}^{M}$. Their entries still have logarithmic height $O(N)$: the residue bound, $H_j \le j$, and $\log D_N = O(N)$ suffice. Replacing r by $r+2$ does not change the asymptotic height estimate. The same counting argument therefore gives a nonzero vector annihilating all three types of coefficients. Its integral is exactly zero, as a formal identity, independently of the arithmetic nature of γ.

Let n_* be the largest index with nonzero coefficient. The pole at $u = -(3n_*)^2$ occurs in R_{n_*} with nonzero residue and in none of the smaller-index kernels. Hence $\sum c_n R_n$ is not the zero rational function. It is continuous on $(0,\infty)$ and has zero integral against a strictly positive measure with full support. It must take both signs there. This proves the last assertion. □

13.1 A fixed-degree positive construction from two independent pairs

Let $\mathcal{K}_N = \ker_{\mathbb{Z}} V$ and define the integer coefficient map

$$\Phi_N(c) = \Big(D_N \sum c_n 16^n,\ D_N \sum c_n p_n\Big).$$

For $c \in \mathcal{K}_N$, the corresponding form is $F_c(\gamma) = Q_c\gamma - P_c$. Define $\sigma_2(N)$ as the least value of $\max(\|c^{(1)}\|_\infty, \|c^{(2)}\|_\infty)$ among pairs in $\mathcal{K}_N$ with

$$\det\begin{pmatrix} Q_1 & P_1 \\ Q_2 & P_2 \end{pmatrix} \neq 0.$$

Set $\sigma_2(N) = \infty$ if no such pair exists. This definition is a property of explicitly given integer and rational matrices. It does not compare any computed value to γ.

Proposition 13.3 (A positive quadratic certificate and its exact missing bound)**.** *For every pair with nonzero determinant, the integer polynomial*

$$W_N(X) = (Q_1X - P_1)^2 + (Q_2X - P_2)^2$$

is strictly positive for every real X. If, along an infinite sequence of indices tending to infinity,

$$\limsup \frac{\log \sigma_2(N)}{N} < \delta_*, \tag{85}$$

then, choosing a pair attaining $\sigma_2(N)$ at each of those indices, the resulting polynomials satisfy $W_N(\gamma) \to 0$, and γ is irrational. No estimate (85) is proved here.

Proof. Two independent coefficient pairs cannot vanish at the same real argument. Equivalently the leading coefficient is positive and the discriminant of W_N is

$$-4(Q_1P_2 - Q_2P_1)^2 < 0.$$

For a pair attaining $\sigma_2(N)$, the raw estimates give

$$|F_{c^{(i)}}(\gamma)| \le D_N d\sigma_2(N) U_N.$$

The logarithm of this upper bound is $-\delta_* N + \log \sigma_2(N) + o(N)$. Under (85) both forms tend to zero, and so does their sum of squares. If $\gamma = a/b$ were rational, then

$$b^2 W_N(\gamma) = (aQ_1 - bP_1)^2 + (aQ_2 - bP_2)^2 \in \mathbb{Z}_{>0},$$

contradicting convergence to zero. The degree is the fixed number 2, so this integerization introduces the fixed factor b^2 only. □

The preceding construction separates positivity from the sign of an integral kernel. It replaces that sign problem by an exact determinant test. It does not supply cheap independent pairs: the counting theorem controls a short null vector, which can lie in $\ker \Phi_N$.

Proposition 13.4 (Why dimension and entry height alone do not bound the pair)**.** *For integers $H \ge 2$, $s \ge 1$, and any number of free coordinates $z_1, \dots, z_k$ with $k \ge 1$, impose*

$$x_i - Hx_{i+1} = 0, \quad y_i - Hy_{i+1} = 0 \quad (0 \le i < s), \qquad \Phi(x, y, z) = (x_s, y_s).$$

The constraint matrix has rank $2s$, entries of absolute value at most H, and arbitrarily many independent null vectors of height 1 with $\Phi = 0$. Appending the two rows of Φ raises its rank by 2. Nevertheless the least height of two null vectors with independent Φ-images is exactly H^s.

Proof. Every null vector has $x_i = H^{s-i}x_s$ and $y_i = H^{s-i}y_s$. If its height is less than H^s, both x_s and y_s are zero. The free coordinate vectors have height 1, and the two choices $(x_s, y_s) = (1, 0), (0, 1)$ give independent images at height exactly H^s. The rank assertions follow directly from the independent chain constraints. □

This is not a failure theorem for the specific proper-kernel matrix. It proves that its additional arithmetic structure must be used: dimension, entry size, and formal rank alone cannot upgrade the counting theorem to the required independent-pair estimate.

14 An actual-matrix height barrier for independent signed outputs

The counting theorem produces short vectors in the integer kernel of the prime-log matrix, but it does not provide two independent coefficient pairs. The following additional synthesis result shows that this distinction is unavoidable for the actual matrices considered here: subexponential-height log-null vectors can have formal outputs of rank at most one. The proof combines the common gamma-coefficient factor with the integral estimate already established above. The underlying integer determinant argument and Smith normal form are standard tools; no historical-priority claim is made for them.

Lemma 14.1 (An integer-output determinant bound)**.** *Let $\Lambda \subseteq \mathbb{Z}^d$ be a lattice, let $a, b : \Lambda \to \mathbb{Z}$ be homomorphisms, and let θ be real. Suppose that, for constants $A, E > 0$ and every $c \in \Lambda$,*

$$|a(c)| \le A\|c\|_\infty, \qquad |\theta a(c) - b(c)| \le E\|c\|_\infty.$$

Put $\mathcal{L} = \{(a(c), b(c)) : c \in \Lambda\} \subseteq \mathbb{Z}^2$. If $\mathcal{L}$ has rank two, let $j = [\mathbb{Z}^2 : \mathcal{L}]$. Then every pair with independent outputs satisfies

$$\|c_1\|_\infty \|c_2\|_\infty \ge \frac{j}{2AE}. \tag{86}$$

In particular, if $2AEK^2 < 1$, all outputs with $\|c\|_\infty \le K$ span a space of dimension at most one, whether or not the full image has rank two.

When the full image has rank two, j is the greatest common divisor of the determinants of all pairs of image vectors. Thus it concerns the actual image lattice, which need not be saturated in $\mathbb{Z}^2$.

Proof. Set $e_i = \theta a(c_i) - b(c_i)$. Cancellation of θ gives

$$a(c_1)b(c_2) - a(c_2)b(c_1) = a(c_2)e_1 - a(c_1)e_2.$$

Its absolute value is at most $2AE\|c_1\|_\infty\|c_2\|_\infty$. A nonzero determinant of two vectors of $\mathcal{L}$ is a nonzero integer multiple of the covolume j, proving (86).

For completeness, a rank-two sublattice of $\mathbb{Z}^2$ has a basis $(a_0, b_0), (0, b_1)$ with $a_0, b_1 > 0$: choose a_0 as the positive generator of its first-coordinate image, choose a vector above it, and choose b_1 as the positive generator of its vertical kernel. Subtracting a multiple of the first vector proves generation by these two vectors. The index and the absolute basis determinant are both $a_0 b_1$. Every pair determinant is a multiple of this number, and the basis pair attains it, proving the gcd assertion.

Finally, if $2AEK^2 < 1$, every determinant of two outputs in the specified height ball is an integer of absolute value less than one, and therefore is zero. If one such output is nonzero, every other one is collinear with it; if none is nonzero, the assertion is immediate. □

Lemma 14.2 (The exact image index and the fibres of the output map)**.** *Let $C \in \mathbb{Z}^{r\times d}$ have row rank r, let $B \in \mathbb{Z}^{2\times d}$, and suppose $\operatorname{rank}\binom{C}{B} = r + 2$. For a full-row-rank integer matrix T with s rows, write $\Delta_s(T)$ for the positive gcd of its s-by-s minors; use $\Delta_0(C) = 1$ when $r = 0$. Then*

$$\left[\mathbb{Z}^2 : B(\ker_{\mathbb{Z}} C)\right] = \frac{\Delta_{r+2}\left(\binom{C}{B}\right)}{\Delta_r(C)}. \tag{87}$$

For any attained output $y = Bc_0$ with $Cc_0 = 0$, its entire fibre is

$$\{c \in \mathbb{Z}^d : Cc = 0,\ Bc = y\} = c_0 + \ker_{\mathbb{Z}} \begin{pmatrix} C \\ B \end{pmatrix}.$$

Both integer kernels in this statement are saturated in $\mathbb{Z}^d$. This does not assert that the image $B(\ker_{\mathbb{Z}} C)$ is saturated in $\mathbb{Z}^2$.

Proof. Put $\mathcal{H} = \binom{C}{B}\mathbb{Z}^d$, $\mathcal{C} = C\mathbb{Z}^d$, and $\mathcal{L} = B(\ker_{\mathbb{Z}} C)$. Projection to the first r coordinates gives the exact sequence of finite abelian groups

$$0 \longrightarrow \mathbb{Z}^2/\mathcal{L} \longrightarrow \mathbb{Z}^{r+2}/\mathcal{H} \longrightarrow \mathbb{Z}^r/\mathcal{C} \longrightarrow 0.$$

Indeed, a class whose first coordinate lies in $\mathcal{C}$ can be represented by $(0, y)$ after subtracting a vector of $\mathcal{H}$, and $(0, y)$ lies in $\mathcal{H}$ exactly when $y \in \mathcal{L}$. Taking group orders gives the ratio of the two image indices.

For a full-row-rank integer matrix, its image index is the gcd of its maximal minors. To see this, apply integer unimodular row and column operations to obtain Smith normal form. Such operations preserve the image index and the gcd of the maximal minors; on the diagonal form both are the product of the nonzero diagonal entries. This proves (87). The fibre identity follows by subtracting c_0. An integer kernel is saturated because $kTc = 0$, for a nonzero integer k, implies $Tc = 0$ in the torsion-free group $\mathbb{Z}^s$. □

Theorem 14.3 (Independent signed outputs have an exponential height cost)**.** *Use the proper kernels, their rational bounds U_n in (82), and the thin blocks of Theorem 13.2. Thus*

$$\mathcal{K}_N = \ker_{\mathbb{Z}} V, \quad D_N = 4\,\mathrm{lcm}(1,\dots,3M), \quad \delta_* = \log(729/16) - 3.$$

Set the explicit positive rational quantities

$$\mathcal{S}_N = \sum_{n=N}^{M} 16^{n-N} = \frac{16^{M-N+1}-1}{15}, \qquad \mathcal{T}_N = \sum_{n=N}^{M} U_n, \qquad G_N = D_N 16^N.$$

For $c \in \mathcal{K}_N$, write the integer form as

$$F_c(\gamma) = Q_c\gamma - P_c, \qquad a_N(c) = \frac{Q_c}{G_N} = \sum_{n=N}^{M} c_n 16^{n-N} \in \mathbb{Z}.$$

Let $\sigma_2(N)$ be the least possible maximum input height of two vectors whose original coefficient pairs (Q_c, P_c) are independent, and set it to ∞ if no such pair exists. Then, for every N for which the block is defined,

$$\sigma_2(N) \geq \left(2D_N\mathcal{S}_N\mathcal{T}_N\right)^{-1/2}. \tag{88}$$

If the normalized image

$$\mathcal{L}_N = \{(a_N(c), P_c) : c \in \mathcal{K}_N\}$$

has rank two and index j_N in $\mathbb{Z}^2$, the sharper bound is

$$\sigma_2(N) \geq \sqrt{\frac{j_N}{2D_N\mathcal{S}_N\mathcal{T}_N}}. \tag{89}$$

In particular, with the extended-real convention for $\log\infty$,

$$\liminf_{N\to\infty} \frac{\log\sigma_2(N)}{N} \geq \frac{\delta_*}{2} = \frac{\log(729/16)-3}{2} > 0. \tag{90}$$

More strongly, if $K_N \geq 1$ and

$$\limsup_{N\to\infty} \frac{\log K_N}{N} < \frac{\delta_*}{2},$$

then, for all sufficiently large N, the formal outputs of all *$c \in \mathcal{K}_N$ with $\|c\|_\infty \leq K_N$ span a space of dimension at most one. In particular this holds for every subexponential height schedule $K_N = \exp(o(N))$, including the heights produced by the counting argument.*

Proof. The exact formulas preceding (84) give

$$P_c = D_N \sum_{n=N}^{M} c_n p_n \in \mathbb{Z}, \qquad F_c(\gamma) = D_N \sum_{n=N}^{M} c_n I_n.$$

Consequently

$$|a_N(c)| \leq \mathcal{S}_N\|c\|_\infty, \qquad |G_N\gamma a_N(c) - P_c| \leq D_N\mathcal{T}_N\|c\|_\infty.$$

The two original pairs are independent exactly when the normalized pairs are independent, since their first coordinates differ by the same nonzero factor G_N. Apply Lemma 14.1 with

$$\theta = G_N\gamma, \quad A = \mathcal{S}_N, \quad E = D_N\mathcal{T}_N.$$

Every nonzero normalized determinant is an integer, and in the rank-two case it is a nonzero multiple of j_N. This proves both finite bounds; if there is no independent pair, (88) holds by the convention $\sigma_2(N) = \infty$.

The thin-block choice gives $M - N = o(N)$, so

$$\log \mathcal{S}_N = o(N).$$

Since U_n decreases, $U_N \le \mathcal{T}_N \le (M - N + 1)U_N$. The proved asymptotic for U_N therefore gives

$$\log \mathcal{T}_N = -\log(729/16)N + o(N).$$

The already used least-common-multiple asymptotic gives $\log D_N = 3N + o(N)$. Hence

$$2D_N\mathcal{S}_N\mathcal{T}_N = \exp\{-\delta_* N + o(N)\}. \tag{91}$$

Taking logarithms of (88) proves (90). Under the stated hypothesis on K_N, the product of (91) with K_N^2 tends to zero. The final conclusion follows from the last assertion of Lemma 14.1. □

Exact arithmetic and scope. The index j_N, whenever the normalized image has rank two, can be computed without any evaluation of γ. Choose a maximal independent set of rows of the integer log matrix as C, and let the two rows of B be $(16^{n-N})_{n=N}^{M}$ and $(D_N p_n)_{n=N}^{M}$. Then Lemma 14.2 gives its exact value. All vectors with a fixed formal output differ by the saturated integer kernel of the stacked matrix; changing a vector inside that fibre changes neither its formal linear form nor its primitive arithmetic error.

The new conclusion is a lower height bound for the actual proper-kernel matrices, not just for an unrelated model matrix. It does not prove that every short output is zero, and it does not exclude independent outputs at larger heights. In particular, the previously stated sufficient target $\limsup \log \sigma_2(N)/N < \delta_*$ is not contradicted: an exponential-rate window between $\delta_*/2$ and δ_* remains. Neither existence of suitable pairs in that window nor irrationality of γ is established here.

15 The remaining estimates and how the results fit together

The positive and signed constructions address different requirements. The following table records proved statements and the precise estimates that are not supplied by this manuscript. A hypothesis in the last column is not used as if it had been established.

Construction	Proved properties	Remaining requirement
Canonical interval selection	Integer coefficients, exact log cancellation, positive forms, terminating selection, divergent denominators, exact limiting dichotomy	An independent proof that $\widehat{q}_m(r_{m+1} - r_m) \to 0$.

Construction	Proved properties	Remaining requirement
Primitive optimization	An exact finite candidate set; a terminating relative optimizer; a three-fraction exact certificate when its hypotheses hold	A proved zero limit of the interval minima for $\xi = \gamma$.
Positive target realization	Every strict rational lower target has a global positive certificate; complete reduction gives the target's exact primitive pair	Selection of an infinite target sequence with primitive error tending to zero.
Localized squares with the repaired cutoff	Positive integral error of order M^{-1}, exact rational bounds, and removal of the designated prime when M is prime	A global denominator bound $q_M = o(M)$ along a specified infinite sequence. The 3- and 5-power towers fail.
Signed thin-block combinations	All logarithms cancel; integerization is included in the exponentially small upper bound; formal-zero combinations also exist	Nonzero integrated output with the required height, or sufficiently small independent output pairs.

The new results sharpen this map. The positive dyadic degree law is optimal on its specified fixed pole grid, even with unlimited coefficient height. It settles an analytic complexity question but leaves the target denominator uncontrolled. The interval theorem proves that least-denominator selection, true primitive optimization, and relative optimization share a limiting rationality dichotomy. Its Fibonacci counterexample shows that the finite errors of these procedures can nevertheless differ by arbitrarily large factors. Finally, the actual signed matrix requires exponential height for two independent outputs. In the notation of the signed construction,

$$\liminf_{N\to\infty} \frac{\log \sigma_2(N)}{N} \geq \frac{\delta_*}{2},$$

whereas a sufficient estimate for its proposed two-square route would be

$$\limsup_{\substack{N\to\infty \\ N\in\mathcal{N}}} \frac{\log \sigma_2(N)}{N} < \delta_*$$

on an infinite set $\mathcal{N}$. The lower bound does not imply that such a set exists, and it does not contradict the stated sufficient upper target.

There is therefore no transfer of nonvanishing from one family to the smallness estimate of another family. A completed irrationality argument would have to establish the missing property within one specified infinite construction. The present results provide exact constructions, optimization tools, arithmetic exclusions, and complexity laws with which that further problem can be studied.

A Source integration, corrected normalization, and reproducibility

A.1 What was integrated

All ten selected result groups are proved in this manuscript. The canonical family, target interpolation, rationality dichotomy, and Laguerre arithmetic come from source A; global positive realization, effective square tests, and support compression come from source B; nested brackets, finite optimization, interpolation valuations, localization, cutoff repair, and signed cancellation come from source C. The additional synthesis results are the sharp degree law, the general

interval dichotomy, the Fibonacci separation example, the exact repaired prime-power costs, and the actual-matrix independence barrier.

Repeated definitions and one duplicate proof of the half-lcm clearing factor have been consolidated. The monomial moment is

$$\int_0^\infty t^s(t-1)\log t\, e^{-t}\, dt = s!\big(1 + sH_s - s\gamma\big).$$

One redundant displayed formula in source C contained an extra s inside these braces. That typographical error is corrected here. The quadratic matrices, the source B moments, and the localization arguments use the normalization displayed above. For example $A(L_1) = 5$ and $B(L_1) = 2$, so $I(L_1) = 5 - 2\gamma$. The integration does not assume that the superseded expression is valid.

Classical inputs include Binet's integral, digamma duplication, Laguerre and Meixner–Pollaczek identities, the prime number theorem, and Meyer's rational isotropy theorem. Each is identified at its point of use. The new monic-polynomial estimate and the image-index formula include their own proofs from factorization, integration by parts, and integer lattice operations.

A.2 Exact checks and their logical role

The proof of the sharp degree estimate is symbolic. As a separate finite check, the leading coefficients, orthogonality, and norms in its shifted Jacobi calculation were verified in rational arithmetic for $\alpha = 0, 1$ and degrees $0, \dots, 12$. The signed determinant argument and the image-index identity were independently checked against the actual proper-kernel coefficient map.

The short Python program below uses only the standard library and exact fractions. It verifies the seven displayed finite error bounds, membership of their targets in the claimed canonical intervals, and the three-fraction certificate for the exact optimum at $m = 10$. It also checks the two strict rational bounds surrounding that optimum. Its successful output is

```
PASS: rational targets, seven error bounds, and exact m=10 optimum
```

The reported finite candidate counts come from the finite ceiling-Euclidean construction; they are not needed by this verifier or by any asymptotic theorem.

These checks supplement the all-index proofs. In particular they do not prove convergence of the gamma-specific primitive errors. The TeX source contains the entire manuscript and bibliography: there are no external `\input`, `\include`, figure, or BibTeX-file dependencies. A standard TeX installation with the listed packages can compile it using `pdflatex` twice or `latexmk -pdf`.

A.3 A self-contained rational certificate verifier

The following code contains no decimal approximation of γ. Finite decimal strings in the comparison table are parsed as exact rational upper bounds.

```
from fractions import Fraction as F
from math import gcd

def bernoulli(n):
    a = [F(0)] * (n + 1)
    for m in range(n + 1):
        a[m] = F(1, m + 1)
```

```
        for j in range(m, 0, -1):
            a[j-1] = j * (a[j-1] - a[j])
    return a[0]

def mu(j):
    return abs(bernoulli(2*j + 2)) / (2*j + 2)

# Store only the harmonic numbers needed by dyadic targets.
harm, h = {}, F(0)
for n in range(1, 2**14 + 1):
    h += F(1, n)
    if n & (n-1) == 0:
        harm[n] = h

def target(m):
    p = [1]
    for j in range(1, m+1):
        t = [0] * (len(p)+1)
        for i, v in enumerate(p):
            t[i] -= v
            t[i+1] += 4**j * v
        p = t
    a = sum(p)
    s = F(sum(i*v for i, v in enumerate(p)), a)
    c = [F(0)] * (len(p)+1)
    for i, v in enumerate(p):
        c[i] += (1+s) * F(v, a)
        c[i+1] -= s * F(v, a)
    assert sum(c) == 1
    assert sum(i*v for i, v in enumerate(c)) == 0
    return sum(v*(harm[2**i]-F(1, 2**(i+1)))
               for i, v in enumerate(c))

r = [target(m) for m in range(14)]
assert all(r[i] < r[i+1] for i in range(13))
rho = lambda m: F(9*(m+1)**2, 16*4**(m+1))
upper = lambda m: r[m] + (r[m+1]-r[m])/(1-rho(m))

# A rational Binet bracket, with an explicit positive remainder.
ell_lo = 2*sum(F(1, (2*j+1)*3**(2*j+1))
               for j in range(110))
ell_hi = ell_lo + F(2, 221*3**221)*F(9, 8)
s = sum((-1)**j * mu(j)/64**(2*j+2) for j in range(20))
rem = mu(20)/64**42
g_lo = harm[64]-F(1, 128)-6*ell_hi+s
g_hi = harm[64]-F(1, 128)-6*ell_lo+s+rem
assert g_lo < g_hi

rows = [
    (0, 4, 7, '4.05096544e-2'),
    (1, 15, 26, '7.60728744e-3'),
    (2, 157, 272, '2.66085322e-3'),
    (4, 33841, 58628, '1.84706856e-6'),
    (6, 9062634, 15700603, '1.53583719e-9'),
    (8, 8788062966751, 15224921118955, '9.04971062e-14'),
    (10, 2323006970406828281, 4024504377931443899,
     '3.81603071e-19'),
]
for m, p, q, bound in rows:
    assert gcd(p, q) == 1
    assert r[m] <= F(p, q) <= r[m+1]
    assert 0 < q*g_lo-p < q*g_hi-p < F(bound)
```

```
p, q = 2323006970406828281, 4024504377931443899
pc, qc = 1055624077073011532, 1828820909171081307
assert q*pc-p*qc == 1
mediant = F(p+pc, q+qc)
assert r[11] < mediant < r[12]
assert upper(12) < F(pc, qc)
assert F(38160307067862345768, 10**38) < q*g_lo-p
assert q*g_hi-p < F(38160307067862345769, 10**38)
print('PASS: rational targets, seven error bounds, and exact m=10 optimum')
```

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